Parameter-free Clipped Gradient Descent Meets Polyak

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Abstract

Gradient descent and its variants are de facto standard algorithms for training machine learning models. As gradient descent is sensitive to its hyperparameters, we need to tune the hyperparameters carefully using a grid search. However, the method is time-consuming, particularly when multiple hyperparameters exist. Therefore, recent studies have analyzed parameter-free methods that adjust the hyperparameters on the fly. However, the existing work is limited to investigations of parameter-free methods for the stepsize, and parameter-free methods for other hyperparameters have not been explored. For instance, although the gradient clipping threshold is a crucial hyperparameter in addition to the stepsize for preventing gradient explosion issues, none of the existing studies have investigated parameter-free methods for clipped gradient descent. Therefore, in this study, we investigate the parameter-free methods for clipped gradient descent. Specifically, we propose Inexact Polyak Stepsize, which converges to the optimal solution without any hyperparameters tuning, and its convergence rate is asymptotically independent of L under L-smooth and (L_0, L_1)-smooth assumptions of the loss function, similar to that of clipped gradient descent with well-tuned hyperparameters. We numerically validated our convergence results using a synthetic function and demonstrated the effectiveness of our proposed methods using LSTM, Nano-GPT, and T5.

1 Introduction

We consider the convex optimization problem:

$$\min_{\boldsymbol{x} \in \mathbb{R}^d} f(\boldsymbol{x}), \tag{1}$$

where the loss function f is convex and lower bounded. In this setting, gradient descent and its variants (Duchi et al., 2011; Kingma and Ba, 2015) are the de facto standard algorithms to minimize the loss function. The performance of the algorithm is highly sensitive to the hyperparameter settings, necessitating the careful tuning of the hyperparameters to achieve best performance. More specifically, when the loss function is L-smooth, gradient descent can achieve the optimal convergence rate $\mathcal{O}(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T})$ when we set the stepsize to $\frac{1}{L}$ where $\boldsymbol{x}_0$ is the initial parameter and $\boldsymbol{x}^\star$ is the optimal solution (Nesterov, 2018). Unfortunately, parameter L is problem-specific and unavailable in practice. Thus, gradient descent must be executed in many times with different hyperparameters to identify the good hyperparameter settings, which is a very time-consuming process. Notably, when multiple hyperparameters are under consideration, this hyperparameter search becomes computationally more demanding.

Several recent studies have examined *parameter-free methods* for tuning hyperparameters on the fly (Berrada et al., 2020; Defazio and Mishchenko, 2023; Orvieto et al., 2022; Jiang and Stich, 2023; Ivgi et al., 2023; Khaled et al., 2023; Orabona and Tommasi, 2017; Carmon and Hinder, 2022).[1]

[1]We use *parameter-free methods* to describe algorithms that provably converge to the optimal solution without any problem-specific parameters for setting their stepsizes and other hyperparameters.

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无参数裁剪梯度下降与Polyak

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摘要

梯度下降及其变体是训练机器学习模型的实际标准算法。由于梯度下降对其超参数敏感，我们需要使用网格搜索仔细调整超参数。然而，该方法耗时，尤其是在存在多个超参数时。因此，最近的研究分析了动态调整超参数的无参数方法。然而，现有工作仅限于对步长的无参数方法的研究，而对其他超参数的无参数方法尚未探索。例如，尽管梯度裁剪阈值除了步长之外是防止梯度爆炸问题的关键超参数，但现有研究均未研究过裁剪梯度下降的无参数方法。因此，在本研究中，我们研究了裁剪梯度下降的无参数方法。具体而言，我们提出了非精确Polyak步长，它在不进行任何超参数调整的情况下收敛到最优解，并且在损失函数满足 L-平滑和(L_0, L_1)-平滑假设时，其收敛速度与裁剪梯度下降在超参数调整良好时的收敛速度相似。我们使用合成函数数值验证了我们的收敛结果，并使用LSTM、Nano-GPT和T5证明了我们提出的方法的有效性。

1 引言

我们考虑凸优化问题：

$$\min_{\boldsymbol{x} \in \mathbb{R}^d} f(\boldsymbol{x}), \tag{1}$$

其中损失函数 f 是凸函数且有下界。在此设置下，梯度下降及其变体（Duchi 等人，2011；Kingma 和 Ba，2015）是事实上的标准算法，用于最小化损失函数。该算法的性能对超参数设置高度敏感，需要仔细调整超参数以获得最佳性能。更具体地说，当损失函数是 L-平滑时，梯度下降在将步长设置为 $\frac{1}{L}$ 时可以实现最优收敛速度 $\mathcal{O}(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T})$，其中 $\boldsymbol{x}_0$ 是初始参数， $\boldsymbol{x}^\star$ 是最优解（Nesterov，2018）。不幸的是，参数 L 是特定于问题的，并且在实践中不可用。因此，梯度下降必须多次执行不同的超参数来识别良好的超参数设置，这是一个非常耗时的过程。值得注意的是，当考虑多个超参数时，这个超参数搜索变得更加计算密集。

近期多项研究已考察无参数方法用于动态调整超参数（Berrada等人，2020；Defazio和Mishchenko，2023；Orvieto等人，2022；Jiang和Stich，2023；Ivgi等人，2023；Khaled等人，2023；Orabona和Tommasi，2017；Carmon和Hinder，2022）。[1]

[1]我们使用<em id='2'>无参数方法来描述那些能够保证收敛到最优解的算法，这些算法在设置步长和其他超参数时无需任何特定问题的参数。

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These methods automatically adjust the stepsize during the training and are guaranteed to converge to the optimal solution without tuning the stepsize. In other words, the stepsize did not require tuning using the grid search. However, the existing parameter-free methods only focus on the stepsize, and parameter-free methods for other hyperparameters have not been explored. For example, in addition to the stepsize, the gradient clipping threshold is an important hyperparameter for training language models (Pascanu et al., 2013; Zhang et al., 2020a,b,c).

Clipped gradient descent can achieve the convergence rate $\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T})$ under the assumption that the loss function is (L_0, L_1)-smooth when we use the optimal stepsize and gradient clipping threshold (Koloskova et al., 2023). In many cases, L_0 is significantly smaller than L (Zhang et al., 2020b). Thus, by comparing with the convergence rate of gradient descent $\mathcal{O}(\frac{L\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T})$, gradient clipping often allows gradient descent to converge faster. However, we must carefully tune two hyperparameters, stepsize and gradient clipping threshold, to achieve this convergence rate. If the gradient clipping threshold is too large, the gradient clipping fails to accelerate the convergence. Moreover, if the gradient clipping threshold is too small, gradient clipping deteriorates rather than accelerating the convergence rate. *Can we develop a parameter-free method whose convergence rate is asymptotically independent of L under (L_0, L_1)-smoothness?*

In this study, we investigate a parameter-free method for clipped gradient descent. First, we provide the better convergence rate of Polyak stepsize (Polyak, 1987) under (L_0, L_1)-smoothness. We discover that the convergence rate of Polyak stepsize matches that of clipped gradient descent with well-tuned stepsize and gradient clipping threshold. Although the convergence rate of Polyak stepsize is asymptotically independent of L under (L_0, L_1)-smooth assumption as clipped gradient descent, it still requires the minimum loss value, which is a problem-specific value. Thus, we make Polyak stepsize parameter-free without losing this property under (L_0, L_1)-smoothness by proposing **Inexact Polyak Stepsize**, which converges to the optimal solution without any problem-specific parameters. We numerically evaluated Inexact Polyak Stepsize using a synthetic function and neural networks, validating our theory and demonstrating the effectiveness of Inexact Polyak Stepsize.

2 Preliminary

2.1 Gradient descent & L-smoothness

One of the most fundamental algorithms for solving Eq. (1) represents the gradient descent:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \eta_t \nabla f(\boldsymbol{x}_t),$$

where $\boldsymbol{x}_0 \in \mathbb{R}^d$ is the initial parameter and $\eta_t > 0$ is the stepsize at t-th iteration. To ensure that gradient descent converges to the optimal solution quickly, we must carefully tune the stepsize η_t. When the stepsize is too large, the training collapses. By contrast, when the stepsize is too small, the convergence rate becomes too slow. Thus, we must search for a proper stepsize as the following theorem indicates.

Assumption 1 (L-smoothness)**.** *There exists a constant $L > 0$ that satisfies the following for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$:*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{2}$$

Theorem 1 (Nesterov (2018, Corollary 2.1.2))**.** *Assume that f is convex and L-smooth, and there exists an optimal solution $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$. Then, gradient descent with stepsize $\eta_t = \frac{1}{L}$ satisfies*

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T}\right), \tag{3}$$

where $\bar{\boldsymbol{x}} := \frac{1}{T}\sum_{t=0}^{T-1} \boldsymbol{x}_t$ and T is the number of iterations.

2.2 Clipped gradient descent & (L_0, L_1)-smoothness

Gradient clipping is widely used to stabilize and accelerate the training of gradient descent (Pascanu et al., 2013; Devlin et al., 2019). Let $c > 0$ be the threshold for gradient clipping. Clipped gradient descent is given by:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \eta_t \min\left\{1, \frac{c}{\|\nabla f(\boldsymbol{x}_t)\|}\right\} \nabla f(\boldsymbol{x}_t). \tag{4}$$

这些方法会自动调整步长并在训练过程中保证收敛到最优解，无需调整步长。换言之，步长无需通过网格搜索进行调优。然而，现有无参数方法仅关注步长，其他超参数的无参数方法尚未被探索。例如，除步长外，梯度裁剪阈值也是训练语言模型的重要超参数（Pascanu等人，2013；Zhang等人，2020a,b,c）。

裁剪梯度下降在假设损失函数为(L_0, L_1)-平滑的情况下，当使用最优步长和梯度裁剪阈值时，可以实现收敛速度 $\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T})$(Koloskova等人，2023)。在许多情况下， L_0 明显小于 L (张等人，2020b)。因此，通过与梯度下降的收敛速度 $\mathcal{O}(\frac{L\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T})$进行比较，梯度裁剪通常使梯度下降收敛得更快。然而，我们必须仔细调整两个超参数，步长和梯度裁剪阈值，才能实现这一收敛速度。如果梯度裁剪阈值过大，梯度裁剪将无法加速收敛。此外，如果梯度裁剪阈值过小，梯度裁剪反而会降低而不是加速收敛速度。我们能开发一种参数无关的方法，其收敛速度渐近地独立于 L 在(L_0, L_1)-平滑性下吗？

在本研究中，我们研究了一种无参数的裁剪梯度下降方法。首先，我们在 (L_0, L_1)-平滑条件下，提供了Polyak步长（Polyak, 1987）更好的收敛率。我们发现Polyak步长的收敛率与具有良好调整步长和梯度裁剪阈值的裁剪梯度下降的收敛率相匹配。尽管在 (L_0, L_1)-平滑假设下，Polyak步长的收敛率与裁剪梯度下降一样，是渐近独立于 L 的，但它仍然需要最小损失值，这是一个特定于问题的值。因此，我们通过提出**非精确**，在 (L_0, L_1)- 平滑条件下，使 Polyak 步长无参数化，同时不失去这一特性

Polyak步长，它在不使用任何特定于问题的参数的情况下收敛到最优解 .
我们使用合成函数和神经网络数值评估了 InexactPolyakStepsize，验证了我们的理论，并展示了非精确Polyak步长的有效性。

2 初步

2.1 梯度下降 & L-平滑性

求解 Eq. (1) 的最基础算法之一是梯度下降:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \eta_t \nabla f(\boldsymbol{x}_t),$$

其中 $\boldsymbol{x}_0 \in \mathbb{R}^d$ 是初始参数， $\eta_t > 0$ 是第 t 次迭代的步长。为了确保梯度下降能够快速收敛到最优解，我们必须仔细调整步长 η_t。当步长过大时，训练会崩溃。相比之下，当步长过小时，收敛速度会变得太慢。因此，我们必须根据以下定理寻找合适的步长。

假设1 (L-平滑性)。 存在一个常数 $L > 0$ ，对所有$\boldsymbol{x},\boldsymbol{y} \in \mathbb{R}^d$ 满足以下条件:

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{2}$$

定理1 (Nesterov (2018, 推论2.1.2))。 假设 是凸的且 L-平滑的，并且存在最优解 $\boldsymbol{x}^\star :=$ $\arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$。那么，步长为 $\eta_t = \frac{1}{L}$ 的梯度下降满足

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T}\right), \tag{3}$$

在 $\bar{\boldsymbol{x}} := \frac{1}{T}\sum_{t=0}^{T-1} \boldsymbol{x}_t$ 处，并且 T 是迭代次数。

2.2 裁剪梯度下降 & (L_0, L_1)-平滑性

梯度裁剪被广泛用于稳定和加速梯度下降（Pascanu等人，2013；Devlin等人，2019）的训练。设 $c > 0$ 为梯度裁剪的阈值。裁剪梯度下降定义为:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \eta_t \min\left\{1, \frac{c}{\|\nabla f(\boldsymbol{x}_t)\|}\right\} \nabla f(\boldsymbol{x}_t). \tag{4}$$

Many prior studies investigated the theoretical benefits of gradient clipping (Koloskova et al., 2023; Zhang et al., 2020a,b,c; Li and Liu, 2022; Sadiev et al., 2023). Zhang et al. (2020b) experimentally found that the gradient Lipschitz constant decreases during the training of various neural networks and is highly correlated with gradient norm $\|\nabla f(\boldsymbol{x})\|$. To describe this phenomenon, Zhang et al. (2020a) introduced a novel smoothness assumption called (L_0, L_1)-smoothness. Then, it has been experimentally demonstrated that the local gradient Lipschitz constant L_0 is thousands of times smaller than the global gradient Lipschitz constant L.

Assumption 2 ((L_0, L_1)-smoothness)**.** *There exists constants $L_0 > 0$ and $L_1 > 0$ that satisfy the following for all $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^d$ with $\|\boldsymbol{x} - \boldsymbol{y}\| \leq \frac{1}{L_1}$:*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq (L_0 + L_1\|\nabla f(\boldsymbol{x})\|)\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{5}$$

Note that (L_0, L_1)-smoothness is strictly weaker than L-smoothness because (L_0, L_1)-smoothness covers L-smoothness by taking $L_1 = 0$. Using the (L_0, L_1)-smoothness assumption, the convergence rate of clipped gradient descent was established as follows.

Theorem 2 (Koloskova et al. (2023, Theorem 2.3))**.** *Assume that f is convex, L-smooth, and (L_0, L_1)-smooth, and there exists an optimal solution $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$. Then, clipped gradient descent with $\eta_t = \frac{1}{L_0}$ and $c = \frac{L_0}{L_1}$ satisfies:*

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}\right), \tag{6}$$

where $\bar{\boldsymbol{x}} := \frac{1}{T}\sum_{t=0}^{T-1}\boldsymbol{x}_t$ and T is the number of iterations.

When the number of iterations T is large, the first term $\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T})$ becomes dominant, and the convergence rate of clipped gradient descent is asymptotically independent of L. Gradient clipping allows for the use of a larger stepsize, and thus, gradient descent converges faster because of $L_0 \ll L$. We can interpret $L \simeq L_0 + L_1 \sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|$. The stepsize of gradient descent in Theorem 1 is $\frac{1}{L_0 + L_1\sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|}$, which is typically very small. By comparing with gradient descent, the coefficient multiplied by the gradient of clipped gradient descent in Theorem 2 is $\min\{\frac{1}{L_0}, \frac{1}{L_1\|\nabla f(\boldsymbol{x}_t)\|}\}$, which is larger than $\frac{1}{L_0 + L_1\sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|}$. Specifically, when parameter $\boldsymbol{x}$ is close to the optimal solution $\boldsymbol{x}^\star$ (i.e., $\|\nabla f(\boldsymbol{x})\|$ is small), clipped gradient descent can use a larger stepsize and then reach the optimal solution faster than gradient descent.

2.3 Polyak stepsize

When f is convex, $\boldsymbol{x}_{t+1}$ and $\boldsymbol{x}_t$ generated by gradient descent satisfy $\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 \leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) + \eta_t^2\|\nabla f(\boldsymbol{x}_t)\|^2$. By minimizing the right-hand side, we can derive well-known Polyak stepsize (Polyak, 1987):

$$\eta_t = \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}, \tag{7}$$

where $f^\star := f(\boldsymbol{x}^\star)$. When f is L-smooth, gradient descent with Polyak stepsize converges to the optimal solution as quickly as gradient descent with $\eta_t = \frac{1}{L}$.

Theorem 3 (Hazan and Kakade (2019, Theorem 1))**.** *Assume that f is convex and L-smooth, and there exists an optimal solution $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$. Then, gradient descent with Polyak stepsize Eq. (7) satisfies:*

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T}\right), \tag{8}$$

where $\bar{\boldsymbol{x}} := \frac{1}{T}\sum_{t=0}^{T-1}\boldsymbol{x}_t$ and T is the number of iterations.

In addition to the L-smooth setting, Polyak stepsize is known to cause gradient descent to converge to the optimal solution with the optimal rate among various settings, e.g., non-smooth convex, smooth convex, and strongly convex settings (Hazan and Kakade, 2019).

许多先前研究调查了梯度裁剪的理论优势（Koloskova等人，2023；Zhang等人，2020a,b,c；Li和Liu，2022；Sadiev等人，2023）。Zhang等人（2020b）通过实验发现，在多种神经网络的训练过程中，梯度Lipschitz常数会减小，并且与梯度范数 $\|\nabla f(\boldsymbol{x})\|$高度相关。为了描述这种现象，Zhang等人（2020a）引入了一种新的平滑性假设，称为(L_0, L_1)-平滑性。随后，实验证明局部梯度Lipschitz常数 L_0 比全局梯度Lipschitz常数 L小数千倍。

假设2 ((L_0, L_1)-平滑性)**.** 存在常数 $L_0 > 0$ 和 $L_1 > 0$，对于所有满足 $\|\boldsymbol{x} - \boldsymbol{y}\| \leq \frac{1}{L_1}$ 的 $\boldsymbol{x}$、$\boldsymbol{y} \in \mathbb{R}^d$，它们满足以下条件：

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq (L_0 + L_1\|\nabla f(\boldsymbol{x})\|)\|\boldsymbol{x} - \boldsymbol{y}\|. \tag{5}$$

注意，(L_0, L_1)-平滑性严格弱于 L-平滑性，因为 (L_0, L_1)-平滑性通过取 $L_1 = 0$ 覆盖了 L-平滑性。使用 (L_0, L_1)-平滑性假设，裁剪梯度下降的收敛速度被证明如下。

定理2 (Koloskova等人 (2023, 定理2.3))**.** 假设 f 是凸的、L-平滑的，并且 L_0、L_1)-平滑的，并且存在最优解 $\boldsymbol{x}^\star: = \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$。那么，具有 $\eta_t = \frac{1}{L_0}$ 和 $c = \frac{L_0}{L_1}$ 的裁剪梯度下降满足：

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}\right), \tag{6}$$

在 $\bar{\boldsymbol{x}}: = \frac{1}{T}\sum_{t=0}^{T-1}\boldsymbol{x}_t$ 和 T 是迭代次数的情况下。

当迭代次数 T 较大时，第一项 $\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T})$ 会占主导地位，裁剪梯度下降的收敛速度渐近地与 L 无关。梯度裁剪允许使用更大的步长，因此梯度下降由于 $L_0 \ll L$ 而收敛更快。我们可以解释 $L \simeq L_0 + L_1\sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|$。定理1中梯度下降的步长是 $\frac{1}{L_0 + L_1\sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|}$，这通常非常小。通过与梯度下降比较，定理2中裁剪梯度下降的梯度乘的系数是 $\min\{\frac{1}{L_0}, \frac{1}{L_1\|\nabla f(\boldsymbol{x}_t)\|}\}$，这比 $\frac{1}{L_0 + L_1\sup_{\boldsymbol{x}}\|\nabla f(\boldsymbol{x})\|}$ 更大。具体来说，当参数 $\boldsymbol{x}$ 接近最优解 $\boldsymbol{x}^\star$（即 $\|\nabla f(\boldsymbol{x})\|$ 很小）时，裁剪梯度下降可以使用更大的步长，然后比梯度下降更快地达到最优解。

2.3 Polyak步长

当 f $\boldsymbol{x}_{t+1}$ $\boldsymbol{x}_t$ $\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 \leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 -$ 为凸函数，且由梯度 $2\eta_t$ f $\boldsymbol{x}_t$ $- f$ $\boldsymbol{x}^\star$ $+ \eta_t^2\|\nabla f$ $\boldsymbol{x}_t$ $\|^2$ 下降生成的函数满足 (( ) ( )) ( ) 。通过最小化右侧，我们可以推导出著名的Polyak步长（Polyak，1987）：

$$\eta_t = \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}, \tag{7}$$

其中 $f^\star := f(\boldsymbol{x}^\star)$。当 f为 L-光滑时，使用Polyak步长的梯度下降算法收敛到最优解的速度与使用 $\eta_t = \frac{1}{L}$的梯度下降算法相同。

定理3 (Hazan and Kakade (2019, 定理1))。假设 f为凸函数且 L-光滑，并且存在最优解 $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$。那么，使用*Polyak*步长公式*(7)*的梯度下降算法满足：

$$f(\bar{\boldsymbol{x}}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T}\right), \tag{8}$$

在 $\bar{\boldsymbol{x}}: = \frac{1}{T}\sum_{t=0}^{T-1}\boldsymbol{x}_t$ 和 T 是迭代次数的情况下。

除了 L-平滑设置外，Polyak步长在各种设置（如非平滑凸、平滑凸和强凸设置）中，已知能使梯度下降以最优速率收敛到最优解（Hazan and Kakade,2019）。

3 Improved convergence result of Polyak stepsize

Before proposing a parameter-free method for clipped gradient descent, in this section, we present a new convergence analysis of Polyak stepsize under (L_0, L_1)-smoothness. Surprisingly, our new analysis reveals that Polyak stepsize achieves exactly the same convergence rate as *clipped* gradient descent with appropriate hyperparameters. A bunch of prior studies established the convergence rates of Polyak stepsize, and it is well-known that Polyak stepsize allows gradient descent to converge as fast as the optimal stepsize. However, our theorem finds that Polyak stepsize achieves a faster convergence rate than gradient descent with the optimal stepsize as clipped gradient descent, and none of the existing studies have found this favorable property of Polyak stepsize.

3.1 Connection between Polyak stepsize and clipped gradient descent

Under (L_0, L_1)-smoothness, we can obtain the following results.

Proposition 1. *Assume that f is convex and (L_0, L_1)-smooth. Then, Polyak stepsize Eq. (7) satisfies:*

$$\min\left\{\frac{1}{4L_0}, \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|}\right\} \leq \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{9}$$

Proof. Assumption 2 and Lemma 2 imply

$$\frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2} \geq \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)}.$$

When $\|\nabla f(\boldsymbol{x}_t)\| < \frac{L_0}{L_1}$, Polyak stepsize is bounded from below by $\frac{1}{4L_0}$. When $\|\nabla f(\boldsymbol{x}_t)\| \geq \frac{L_0}{L_1}$, we have

$$\frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)} \geq \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|}.$$

Therefore, we can conclude the statement. □

Under L-smoothness, the lower bound of Polyak stepsize was obtained as follows.

Proposition 2 (Jiang and Stich (2023, Lemma 15))**.** *Assume that f is convex and L-smooth. Then, Polyak stepsize Eq. (7) satisfies:*

$$\frac{1}{2L} \leq \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{10}$$

By comparing Propositions 1 and 2, Proposition 1 shows that Polyak stepsize does not become excessively small when the parameter approaches the optimal solution (i.e., $\|\nabla f(\boldsymbol{x})\|$ approaches zero), similar to clipped gradient descent. If we choose the stepsize and gradient clipping threshold as in Theorem 2, clipped gradient descent can be written as follows:

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \min\left\{\frac{1}{L_0}, \frac{1}{L_1\|\nabla f(\boldsymbol{x}_t)\|}\right\} \nabla f(\boldsymbol{x}_t). \tag{11}$$

Thus, Proposition 1 implies that Polyak stepsize can be regarded as internally estimating the hyperparameters for clipped gradient descent, as shown in Theorem 2.

3.2 Convergence analysis of Polyak stepsize under (L_0, L_1)-smoothness

Based on the relationship between Polyak stepsize and clipped gradient descent in Sec. 3.1, we provide a new convergence result for Polyak stepsize under (L_0, L_1)-smoothness. The proof is deferred to Sec. A.

Theorem 4. *Assume that f is convex, L-smooth, and (L_0, L_1)-smooth, and there exists an optimal solution $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$. Let T be the number of iterations and define $\tau := \arg\min_{0\leq t\leq T-1} f(\boldsymbol{x}_t)$. Then, gradient descent with Polyak stepsize Eq. (7) satisfies:*

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}\right). \tag{12}$$

3 Polyak步长的改进收敛结果

在提出裁剪梯度下降的无参数方法之前，本节首先对Polyak步长在(L_0, L_1)-平滑条件下的收敛性进行了新的分析。令人惊讶的是，我们的新分析揭示Polyak步长与具有适当超参数的裁剪梯度下降具有完全相同的收敛速率。大量先前研究建立了Polyak步长的收敛速率，众所周知，Polyak步长允许梯度下降以最优步长最快的速度收敛。然而，我们的定理发现，在裁剪梯度下降中，Polyak步长比具有最优步长的梯度下降实现了更快的收敛速率，而现有研究均未发现Polyak步长这一有利的特性。

3.1 Polyak步长与裁剪梯度下降之间的联系

在(L_0, L_1)-平滑条件下，我们可以得到以下结果。

命题1。 假设 f 是凸的并且<样式 *id='4'*>*(*</样式>L_0, L_1<样式 *id='8'*>*)*</样式>-平滑。那么，*Polyak*步长公式*(7)*满足:

$$\min\left\{\frac{1}{4L_0}, \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|}\right\} \leq \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{9}$$

证明。 假设2和引理2意味着

$$\frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2} \geq \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)}.$$

当 $\|\nabla f(\boldsymbol{x}_t)\| < \frac{L_0}{L_1}$时， Polyak步长由 $\frac{1}{4L_0}$ 下界约束。当 $\|\nabla f(\boldsymbol{x}_t)\| \geq \frac{L_0}{L_1}$时，我们有 —

$$\frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)} \geq \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|}.$$

因此，我们可以得出该结论。 □

在 L-平滑条件下，Polyak步长的下界如下所示。

命题2 (Jiang和Stich（2023年，引理15）)。 假设 f 是凸的且 L-平滑。那么，*Polyak*步长公式*(7)*满足:

$$\frac{1}{2L} \leq \frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{10}$$

通过比较命题1和命题2，命题1表明当参数接近最优解（即 $\|\nabla f(\boldsymbol{x})\|$ 接近零）时，Polyak步长不会变得过小，这与梯度裁剪下降相似。如果我们选择定理2中所示的步长和梯度裁剪阈值，梯度裁剪下降可以表示如下：

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t - \min\left\{\frac{1}{L_0}, \frac{1}{L_1\|\nabla f(\boldsymbol{x}_t)\|}\right\} \nabla f(\boldsymbol{x}_t). \tag{11}$$

因此，命题1表明Polyak步长可以被视为内部估计梯度裁剪下降的超参数，如定理2所示。

3.2 Polyak步长在(L_0, L_1)-平滑条件下的收敛性分析

基于第3.1节中Polyak步长与裁剪梯度下降的关系，我们为Polyak步长在(L_0, L_1)-平滑条件下提供了一个新的收敛结果。证明推迟到第A节。

定理4。 假设 f 是凸的， L-平滑的，并且<样式*id='5'*>*(*</样式>L_0, L_1<样式*id='9'*>*)*</样式>-平滑的，并且存在一个最优解 $\boldsymbol{x}^\star$<样式*id='12'*>: </样式>$=$ <样式*id='15'*>*arg min*</样式>$_{\boldsymbol{x}\in\mathbb{R}^d}$ f<样式*id='18'*>*(*</样式>$\boldsymbol{x}$<样式*id='21'*>*)*</样式>。令 T 为迭代次数，并定义 τ<样式*id='25'*>: </样式>$=$<样式*id='28'*>*arg min*</样式>$_{0\leq t\leq T-1}$ f<样式*id='31'*>*(*</样式>$\boldsymbol{x}_t$<样式*id='34'*>*)*</样式>。那么，使用*Polyak*步长公式*(7)*的梯度下降满足:

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}\right). \tag{12}$$

By comparing Theorem 4 with Theorem 2, the convergence rate of Polyak stepsize is the same as that of clipped gradient descent. Thus, Polyak stepsize can converge faster than the optimal stepsize given in Theorem 1 when $L_0 \ll L$. Many prior studies analyzed the convergence rate of Polyak stepsize and discussed the relationship between Polyak stepsize and gradient descent with the optimal stepsize (Polyak, 1987; Loizou et al., 2021; Galli et al., 2023; Berrada et al., 2020). However, they only recognized Polyak stepsize as making gradient descent converge with the same convergence rate as the optimal stepsize, and none of the prior studies have found this relationship between Polyak stepsize and clipped gradient descent. Our new convergence result is the first to discover that the Polyak stepsize can achieve the same convergence rate not only as gradient descent with an appropriate stepsize but also as clipped gradient descent with an appropriate stepsize and gradient clipping threshold.

4 Making clipped gradient descent parameter-free

In the previous section, we found that the convergence rate of Polyak stepsize is asymptotically independent of L under (L_0, L_1)-smoothness as clipped gradient descent with appropriate hyper-parameters. However, Polyak stepsize requires the minimum loss value $f^\star$, which is a problem-specific parameter. In this section, we propose a method that can remove the prior knowledge of $f^\star$ from Polyak stepsize without losing the property of asymptotic independence of L under (L_0, L_1)-smoothness.

4.1 Inexact Polyak Stepsize

To make Polyak stepsize parameter-free, several prior studies have proposed the use of lower bound of $f^\star$ instead of $f^\star$ (Loizou et al., 2021; Orvieto et al., 2022; Jiang and Stich, 2023). The loss functions commonly used in machine learning models are non-negative. Thus, the lower bound of $f^\star$ is trivially obtained as zero and is not a problem-specific parameter. By utilizing this lower bound, a straightforward approach to make Polyak stepsize independent of problem-specific parameters is replacing $f^\star$ in Polyak stepsize with the lower bound $l^\star$ as follows:

$$\eta_t = \frac{f(\boldsymbol{x}_t) - l^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{13}$$

However, the stepsize in Eq. (13) becomes excessively large as the parameter approaches the optimal solution, and it does not lead to the optimal solution (Loizou et al., 2021). This is because $\|\nabla f(\boldsymbol{x}_t)\|$ approaches zero, while $f(\boldsymbol{x}_t) - l^\star$ approaches $f^\star - l^\star (> 0)$, which makes the stepsize in Eq. (13) excessively large as the parameter approaches the optimal solution. To mitigate this issue, DecSPS (Orvieto et al., 2022) and AdaSPS (Jiang and Stich, 2023), which are parameter-free methods based on Polyak stepsize that use $l^\star$ instead of $f^\star$, make the stepsize monotonically non-increasing to converge to the optimal solution.

However, making the stepsize monotonically non-increasing loses the fruitful property that the convergence rate of Polyak stepsize is asymptotically independent of L as clipping gradient descent under (L_0, L_1)-smoothness. This is because Polyak stepsize and clipped gradient descent make the convergence rate asymptotically independent of L by increasing the stepsize when the parameter approaches the optimal solution. In fact, we evaluated DecSPS and AdaSPS with a synthetic function in Sec. 6.1, demonstrating that the convergence deteriorates as L increases.

Algorithm 1 Inexact Polyak Stepsize

Input: The number of iterations T and lower bound $l^\star$.
$f^{\text{best}}, \boldsymbol{x}^{\text{best}} \leftarrow f(\boldsymbol{x}_0), \boldsymbol{x}_0$.
for $t = 0, 1, \cdots, T-1$ **do**
  $\boldsymbol{x}_{t+1} \leftarrow \boldsymbol{x}_t - \frac{f(\boldsymbol{x}_t) - l^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2} \nabla f(\boldsymbol{x}_t)$.
  if $f(\boldsymbol{x}_{t+1}) \leq f^{\text{best}}$ **then**
    $f^{\text{best}}, \boldsymbol{x}^{\text{best}} \leftarrow f(\boldsymbol{x}_{t+1}), \boldsymbol{x}_{t+1}$.
return $\boldsymbol{x}^{\text{best}}$.

通过比较定理4与定理2，Polyak步长的收敛速度与裁剪梯度下降相同。因此，当 $L_0 \ll L$ 时，Polyak步长可以比定理1中给出的最优步长收敛得更快。许多先前研究分析了Polyak步长的收敛速度，并讨论了Polyak步长与使用最优步长的梯度下降之间的关系（Polyak, 1987；Loizou等人，2021；Galli等人，2023；Berrada等人，2020）。然而，它们只认识到Polyak步长使梯度下降的收敛速度与最优步长相同，并且先前研究都没有发现Polyak步长与裁剪梯度下降之间的关系。我们的新收敛结果首次发现，Polyak步长不仅可以像使用适当步长的梯度下降一样实现相同的收敛速度，还可以像使用适当步长和梯度裁剪阈值的裁剪梯度下降一样实现相同的收敛速度。

4 裁剪梯度下降的无参数化

在上一节中，我们发现Polyak步长的收敛速度在(L_0, L_1)-平滑条件下渐近独立于 L，这与具有适当超参数的裁剪梯度下降相同。然而，Polyak步长需要最小损失值 $f^\star$，这是一个特定于问题的参数。在本节中，我们提出一种方法，可以在不损失 L 在(L_0, L_1)-平滑条件下渐近独立性的前提下，从Polyak步长中移除对 $f^\star$ 的先验知识。

4.1 非精确Polyak步长

为了使Polyak步长无参数化，多项先验研究提出了使用 $f^\star$的下界代替 $f^\star$(Loizou等人，2021；Orvieto等人，2022；Jiang和Stich，2023)。机器学习模型中常用的损失函数是非负的。因此，$f^\star$的下界显然为零，并且不是特定于问题的参数。通过利用这个下界，一种使Polyak步长独立于特定于问题的参数的简单方法是将Polyak步长中的 $f^\star$ 替换为下界 $l^\star$，如下所示：

$$\eta_t = \frac{f(\boldsymbol{x}_t) - l^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}. \tag{13}$$

然而，当参数接近最优解时，式(13)中的步长会变得过大，并且无法得到最优解（Loizou等人，2021）。这是因为 $\|\nabla f(\boldsymbol{x}_t)\|$趋近于零，而 $f(\boldsymbol{x}_t) - l^\star$ 趋近于 $f^\star - l^\star (> 0)$，这使得当参数接近最优解时，式(13)中的步长变得过大。为了解决这个问题，基于Polyak步长且使用 $l^\star$ 代替 $f^\star$的无参数方法DecSPS（Orvieto等人，2022）和AdaSPS（Jiang和Stich，2023）使步长单调非增，从而收敛到最优解。

然而，使步长单调非递增会失去 Polyak 步长收敛速度渐近独立于 L（在 L_0, L_1)-平滑度下的裁剪梯度下降这一有益特性。这是因为 Polyak 步长和裁剪梯度下降通过在参数接近最优解时增加步长，使得收敛速度渐近独立于 L。事实上，我们在第 6.1 节使用合成函数评估了 DecSPS 和 AdaSPS，证明随着 L 的增加，收敛性会恶化。

算法1非精确Polyak步长

Input: The number of iterations T and lower bound $l^\star$.
$f^{\text{best}}, \boldsymbol{x}^{\text{best}} \leftarrow f(\boldsymbol{x}_0), \boldsymbol{x}_0$.
for $t = 0, 1, \cdots, T-1$ **do**
  $\boldsymbol{x}_{t+1} \leftarrow \boldsymbol{x}_t - \frac{f(\boldsymbol{x}_t) - l^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2} \nabla f(\boldsymbol{x}_t)$.
  if $f(\boldsymbol{x}_{t+1}) \leq f^{\text{best}}$ **then**
    $f^{\text{best}}, \boldsymbol{x}^{\text{best}} \leftarrow f(\boldsymbol{x}_{t+1}), \boldsymbol{x}_{t+1}$.
return $\boldsymbol{x}^{\text{best}}$.

To address this issue, we propose **Inexact Polyak Stepsize**, whose details are described in Alg. 1. As discussed above, we cannot make the stepsize decrease to maintain the asymptotic independence of L under (L_0, L_1)-smoothness. Thus, we set the stepsize as follows:

$$\eta_t = \frac{f(\boldsymbol{x}_t) - l^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2}, \tag{14}$$

where T denotes the number of iterations. Instead of making the stepsize decrease, we propose returning the parameter for which the lowest loss is achieved as the final parameter.

4.2 Convergence analysis of Inexact Polyak Stepsize

The following theorem provides the convergence rate of Inexact Polyak Stepsize. The proof is deferred to Sec. B.

Theorem 5. *Assume that f is convex, L-smooth, and (L_0, L_1)-smooth, and there exists an optimal solution $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$. Let T be the number of iterations and $\sigma^2 := f^\star - l^\star$. Then, $\boldsymbol{x}$ generated by Alg. 1 satisfies:*

$$f(\boldsymbol{x}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2 L\sigma^4}{L_0^2 T}\right). \tag{15}$$

Asymptotic independence of L: When the number of iterations T is large, only the first term $\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2+\sigma^2}{\sqrt{T}})$ becomes dominant in the convergence rate, which does not depend on L. Thus, Theorem 5 shows that Inexact Polyak Stepsize successfully inherits the favorable property of Polyak stepsize under (L_0, L_1)-smoothness. In addition to Inexact Polyak Stepsieze, DecSPS (Orvieto et al., 2022) and AdaSPS (Jiang and Stich, 2023) have been proposed as parameter-free methods that use $l^\star$ instead of $f^\star$ in Polyak stepsize. However, these prior methods fail to inherit the favorable property of Polyak stepsize, and their convergence rates deteriorate when L is large because these methods decrease the stepsize during the training. In fact, we evaluated DecSPS and AdaSPS with a synthetic function in Sec. 6.1, demonstrating that convergence rates of DecSPS and AdaSPS are degraded when L becomes large, whereas the convergence rate of Inexact Polyak Stepsize does not depend on L.

Removing dependence on D_T: The convergence rates of DecSPS and AdaSPS depend on $D_T (:= \max_{0\leq t\leq T}\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|)$. Thus, strictly speaking, these convergence rates cannot show that DecSPS and AdaSPS converge to the optimal solution because D_T may increase as the number of iterations T increases. For instance, if D_T increase with $\Omega(T^{\frac{1}{4}})$, the convergence rate of AdaSPS is $\mathcal{O}(L\sigma + L^2)$, which does not show that AdaSPS converges to the optimal solution. In contrast, the convergence rate in Eq. (15) depends on only $\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|$. Theorem 5 indicates that Inexact Polyak Stepsize converges to the optimal solution.

Convergence rate with respect to T: Inexact Polyak Stepsize successfully achieves the asymptotic independence of L, while it slows down the convergence rate with respect to the number of iterations T by comparing clipped gradient descent with proper hyperparameters. The convergence rate of Inexact Polyak Stepsize $\mathcal{O}(\frac{L_0}{\sqrt{T}})$ is not optimal in terms of T, and there may be room to improve this rate. For instance, the adaptive methods proposed by Hazan and Kakade (2019) might be used to

Table 1: Summary of convergence rates of parameter-free methods based on Polyak stepsize. All convergence results are the ones under convex, L-smoothness, and (L_0, L_1)-smoothness. We define $D_T := \max_{0\leq t\leq T}\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|$.

Algorithm	Convergence Rate	Assumption
DecSPS (Orvieto et al., 2022)[(a)]	$\mathcal{O}\left(\frac{\max\{L,\eta_0^{-1}\}D_T^2+\sigma^2}{\sqrt{T}}\right)$	1
AdaSPS (Jiang and Stich, 2023)[(a)]	$\mathcal{O}\left(\frac{LD_T^2\sigma}{\sqrt{T}} + \frac{L^2D_T^4}{T}\right)$	1
Inexact Polyak Stepsize (This work)	$\mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2+\sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2L\sigma^4}{L_0^2T}\right)$	1, 2[(b)]

(a) We present the convergence rates of DecSPS and AdaSPS in the deterministic setting to compare DecSPS, AdaSPS, and Inexact Polyak Stepsize in the same deterministic setting, while Orvieto et al. (2022) and Jiang and Stich (2023) also analyzed the rate rates in the stochastic setting.

(b) If f is L-smooth, f is (L_0, L_1)-smooth because (L_0, L_1)-smoothness assumption is strictly weaker than L-smoothness assumption.

为解决此问题，我们提出 **非精确Polyak步长**，其细节在算法1中描述。如前所述，我们无法使步长减小以保持L 在(L_0, L_1)-光滑性下的渐近独立性。因此，我们按如下方式设置步长：

$$\eta_t = \frac{f(\boldsymbol{x}_t) - l^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2}, \tag{14}$$

其中 T 表示迭代次数。我们不是让步长减小，而是提出返回达到最低损失时的参数作为最终参数。

4.2 非精确Polyak步长的收敛分析

以下定理提供了非精确Polyak步长的收敛速度。证明被推迟到B节。

定理5。假设 f 是凸的，L-平滑的，并且 (L_0, L_1)-平滑的，并且存在一个最优解 $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}\in\mathbb{R}^d} f(\boldsymbol{x})$。令 T 是迭代次数和 $\sigma^2 := f^\star - l^\star$。然后，由算法*1*生成的 $\boldsymbol{x}$ 满足：

$$f(\boldsymbol{x}) - f(\boldsymbol{x}^\star) \leq \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2 L\sigma^4}{L_0^2 T}\right). \tag{15}$$

渐近独立性： L**：**当迭代次数 T 较大时，收敛速度中仅第一项$\mathcal{O}(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2+\sigma^2}{\sqrt{T}})$起主导作用，且~~与 L 无关。因~~此，定理5表明，非精确Polyak步长在 (L_0, L_1)-平滑条件下，成功继承了Polyak步长的优良性质。除了非精确Polyak步长，DecSPS (Orvieto等人, 2023) 和 AdaSPS (Jiang和Stich, 2023) 也被提出作为无参数方法，它们在Polyak步长中使用 $l^\star$ 而非 $f^\star$。然而，这些先前方法未能继承Polyak步长的优良性质，且当 L 较大时，它们的收敛速度会变差，因为这些方法在训练过程中会减小步长。事实上，我们在Sec.6.1中使用合成函数评估了DecSPS和AdaSPS，结果表明当L 变得较大时，DecSPS和AdaSPS的收敛速度会变差，而非精确Polyak步长的收敛速度则与 L无关。

去除对 D_T的依赖：DecSPS和AdaSPS的收敛速度取决于 $D_T(:=\max_{0\leq t\leq T}\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|)$。因此，严格来说，这些收敛速度不能证明DecSPS和AdaSPS收敛到最优解，因为 D_T 可能会随着迭代次数 T的增加而增大。例如，如果 D_T随着$\Omega(T^{\frac{1}{4}})$的增加而增大，那么AdaSPS的收敛速度是 $\mathcal{O}(L\sigma + L^2)$，这并不能证明AdaSPS收敛到最优解。相比之下，公式(15)中的收敛速度仅取决于 $\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|$。定理5表明，非精确Polyak步长收敛到最优解。

相对于 T：非精确Polyak步长成功实现了 L的渐近独立性，但在与具有适当超参数的裁剪梯度下降进行比较时，它减缓了相对于迭代次数T 的收敛速度。非精确Polyak步长 $\mathcal{O}(\frac{L_0}{\sqrt{T}})$ 在 T方面并非最优，并且可能存在改进此速度的空间。例如，可以采用Hazan和Kakade（2019）提出的自适应方法。

表1：基于Polyak步长的无参数方法收敛速度汇总。所有收敛结果都是在凸、L-平滑和(L_0, L_1)-平滑条件下得出的。我们定义$D_T := \max_{0\leq t\leq T}\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|$。

算法	收敛率	假设
DecSPS（Orvieto等人，2022）[(a)]	$\mathcal{O}\left(\frac{\max\{L,\eta_0^{-1}\}D_T^2+\sigma^2}{\sqrt{T}}\right)$	1
AdaSPS（Jiang和Stich，2023）[(a)]	$\mathcal{O}\left(\frac{LD_T^2\sigma}{\sqrt{T}} + \frac{L^2D_T^4}{T}\right)$	1
非精确Polyak步长（本文工作）	$\mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2+\sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2L\sigma^4}{L_0^2T}\right)$	1, 2[(b)]

(a) 我们在确定性设置中展示了DecSPS和AdaSPS的收敛速度，以在相同的确定性设置中比较DecSPS、AdaSPS和非精确Polyak步长，而Orvieto等人(2022)和Jiang和Stich(2023)也在随机设置中分析了收敛速度。(b) 如果 f 是 L-平滑的，f 是(L_0, L_1)-平滑的，因为(L_0, L_1)-平滑性假设严格弱于 L-平滑性假设。

alleviate this issue. However, the parameter-free methods for clipped gradient descent have not been explored well in the existing studies. We believe that Inexact Polyak Stepsize is the important first step for developing parameter-free clipped gradient descent.

5 Related work

Gradient clipping: Gradient clipping was initially proposed to mitigate the gradient explosion problem for training RNN and LSTM (Mikolov et al., 2010; Merity et al., 2018) and is now widely used to accelerate and stabilize the training not only for RNN and LSTM, but also for various machine learning models, especially language models (Devlin et al., 2019; Raffel et al., 2019). Recently, many studies have investigated the theoretical benefits of gradient clipping and analyzed the convergence rate of clipped gradient descent under (1) (L_0, L_1)-smoothness assumption (Koloskova et al., 2023; Zhang et al., 2020a,b) and (2) heavy-tailed noise assumption (Zhang et al., 2020c; Li and Liu, 2022; Sadiev et al., 2023). (1) Zhang et al. (2020b) found that the local gradient Lipschitz constant is correlated with the gradient norm. To describe this phenomenon, Zhang et al. (2020b), Zhang et al. (2020a), and Koloskova et al. (2023) introduced the new assumption, (L_0, L_1)-smoothness, providing the convergence rate of clipped gradient descent under (L_0, L_1)-smoothness. Then, they showed that gradient clipping can improve the convergence rate of gradient descent, as we introduced in Sec. 2.2. (2) Besides (L_0, L_1)-smoothness, Zhang et al. (2020c) pointed out that the distribution of stochastic gradient noise is heavy-tailed for language models. Then, it has been shown that gradient clipping can make the stochastic gradient descent robust against the heavy-tailed noise of stochastic gradient (Li and Liu, 2022; Sadiev et al., 2023; Zhang et al., 2020c).

Parameter-free methods: Hyperparameter-tuning is one of the most time-consuming tasks for training machine learning models. To alleviate this issue, many parameter-free methods that adjust the stepsize on the fly have been proposed, e.g., Polyak-based stepsize (Berrada et al., 2020; Hazan and Kakade, 2019; Loizou et al., 2021; Mukherjee et al., 2023; Orvieto et al., 2022; Jiang and Stich, 2023), AdaGrad-based methods (Ivgi et al., 2023; Khaled et al., 2023), and Dual Averaging-based methods (Orabona and Tommasi, 2017; Defazio and Mishchenko, 2023). However, parameter-free methods for hyperparameters, except for stepsizes, have not been studied. In this work, we studied the parameter-free methods for two hyperparameters, the stepsize and gradient clipping threshold, and then proposed Inexact Polyak Stepsize, which converges to the optimal solution without tuning any hyperparameters and its convergence rate is asymptotically independent of L as clipped gradient descent with well-tuned hyperparameters.

6 Numerical evaluation

In this section, we evaluate our theory numerically. In Sec. 6.1, we evaluate Polyak stepsize and Inexact Polyak Stepsize using a synthetic function, varying that their convergence rates are asymptotically independent of L. In Sec. 6.2, we show the results obtained using neural networks.

6.1 Synthetic function

Setting: In this section, we validate our theory for Polyak stepsize and Inexact Polyak Stepsize using a synthetic function. We set the loss function as $f(x) = \frac{L_0 L_1^2}{72} x^4 + \frac{L_0}{4} x^2 + f^\star$, which is (L_0, L_1)-smooth for any $L_0 > 0$ and $L_1 > 0$ (See Proposition 3 in Appendix). We set L_0 to 1, $\boldsymbol{x}_0$ to 5, $f^\star = 1$, and $l^\star = 0$ and then evaluated various methods when varying L_1.

Results: We show the results in Fig. 1. The results indicate that gradient descent converges slowly when L_1 is large, whereas Polyak stepsize and clipped gradient descent does not depend on L_1. These observations are consistent with those discussed in Sec. 3, which shows that the convergence rate of Polyak stepsize is asymptotically independent of L as in clipped gradient descent. By comparing DecSPS, AdaSPS, and Inexact Polyak Stepsize, which are parameter-free methods, the convergence rates of DecSPS and AdaSPS degrade as L_1 increases. Thus, DecSPS and AdaSPS lose the favorable property of asymptotic independence of L under (L_0, L_1)-smoothness. In contrast, the convergence behavior of Inexact Polyak Stepsize does not depend on L_1, which is consistent with Theorem 5, and Inexact Polyak Stepsize successfully inherits the Polyak stepsize under (L_0, L_1)-smoothness.

缓解这个问题。然而，现有的研究尚未充分探索用于裁剪梯度下降的无参数方法。我们相信，非精确Polyak步长是开发无参数裁剪梯度下降的重要第一步。

5 相关工作

梯度裁剪： 梯度裁剪最初是为了缓解训练循环神经网络（RNN）和长短期记忆网络（LSTM）时的梯度爆炸问题而提出的（Mikolov等人，2010；Merity等人，2018），现在被广泛用于加速和稳定训练，不仅适用于RNN和LSTM，也适用于各种机器学习模型，尤其是语言模型（Devlin等人，2019；Raffel等人，2019）。最近，许多研究调查了梯度裁剪的理论优势，并分析了裁剪梯度下降在（1）(L_0, L_1)-平滑假设下（Koloskova等人，2023；Zhang等人，2020a,b）和（2）重尾噪声假设下的收敛速度（Zhang等人，2020c；Li和Liu，2022；Sadiev等人，2023）。（1）Zhang等人（2020b）发现局部梯度Lipschitz常数与梯度范数相关。为了描述这种现象，Zhang等人（2020b）、Zhang等人（2020a）和Koloskova等人（2023）引入了新的假设，(L_0, L_1)-平滑，提供了裁剪梯度下降在(L_0, L_1)-平滑下的收敛速度。然后，他们表明梯度裁剪可以提高梯度下降的收敛速度，正如我们在第2.2节中介绍的那样。（2）除了(L_0, L_1)-平滑外，Zhang等人（2020c）指出语言模型中的随机梯度噪声分布是重尾的。然后，已经表明梯度裁剪可以使随机梯度下降对随机梯度的重尾噪声具有鲁棒性（Li和Liu，2022；Sadiev等人，2023；Zhang等人，2020c）。

无参数方法： 超参数调优是训练机器学习模型中最耗时的一项任务。为缓解这一问题，许多动态调整步长的无参数方法已被提出，例如基于Polyak的步长（Berrada 等人，2020；Hazan 和 Kakade, 2019；Loizou 等人，2021；Mukherjee 等人，2023；Orvieto 等人，2022；Jiang 和 Stich, 2023）、基于AdaGrad的方法（Ivgi 等人，2023；Khaled 等人，2023）以及基于Dual Averaging的方法（Orabona 和 Tommasi，2017；Defazio 和 Mishchenko，2023）。然而，除了步长之外，用于其他超参数的无参数方法尚未得到研究。在本工作中，我们研究了用于步长和梯度裁剪阈值的两种无参数方法，并提出了非精确Polyak步长，该方法无需调优任何超参数即可收敛到最优解，且其收敛速度渐近独立于 L，与超参数调整良好的裁剪梯度下降方法相当。

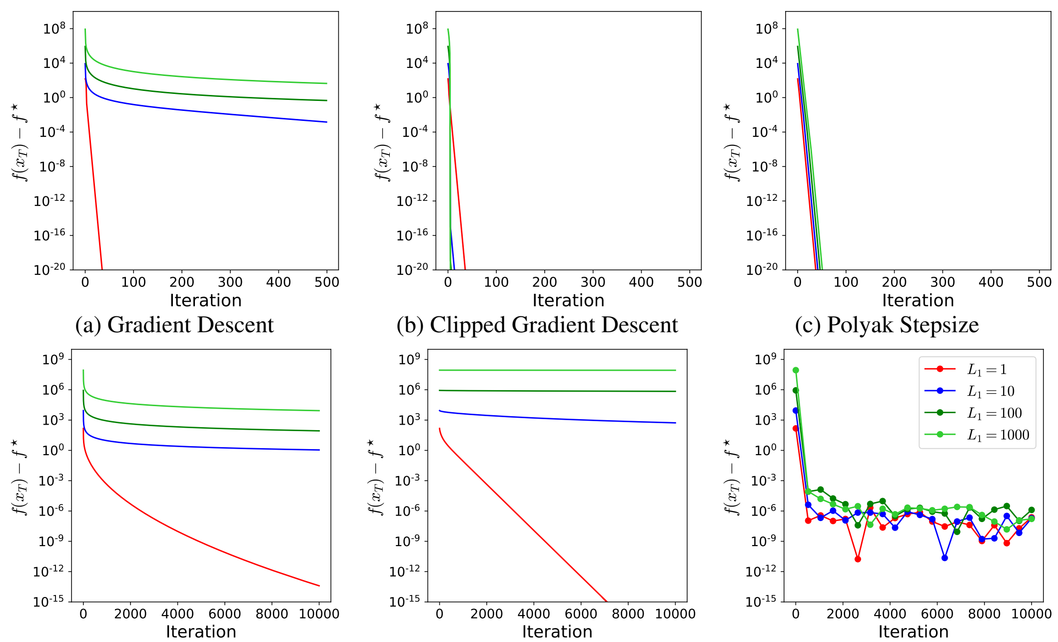
6 数值评估

在本节中，我们通过数值方法评估我们的理论。在 6.1 节中，我们使用合成函数评估 Polyak 步长和非精确 Polyak 步长，并验证它们的收敛速度渐近独立于 L。在 6.2 节中，我们展示了使用神经网络得到的结果。

6.1 合成函数

设置： 在本节中，我们使用合成函数验证 Polyak 步长和非精确 Polyak 步长的理论。我们将损失函数设置为 $f(x) = \frac{L_0 L_1^2}{72} x^4 + \frac{L_0}{4} x^2 + f^\star$，它在任何 $L_0 > 0$ 和 $L_1 > 0$ 下都是 (L_0, L_1)-光滑的（参见附录中的命题3）。我们将 L_0 设置为 1，$\boldsymbol{x}_0$ 设置为 5，$f^\star = 1$，$l^\star = 0$，然后评估了当 L_1 变化时各种方法的性能。

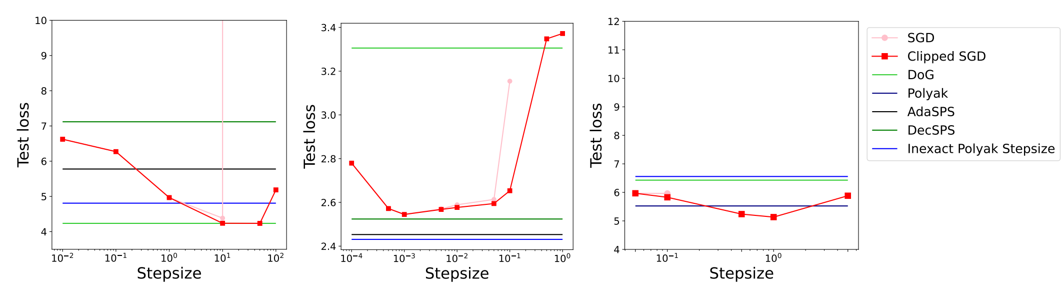
结果： 我们在图1中展示了结果。结果表明，当 L_1 较大时，梯度下降收敛缓慢，而Polyak步长和裁剪梯度下降不依赖于 L_1。这些观察结果与第3节中讨论的内容一致，该节表明 Polyak步长的收敛速度渐近地独立于 L，就像裁剪梯度下降一样。通过比较DecSPS、AdaSPS 和非精确Polyak步长，这些是无参数方法，DecSPS 和 AdaSPS 的收敛速度随着 L_1 的增加而下降。因此，DecSPS 和 AdaSPS 在 (L_0, L_1)-平滑条件下失去了 L 渐近独立的有利特性。相比之下，非精确Polyak步长的收敛行为不依赖于 L_1，这与定理5一致，并且非精确Polyak步长成功地继承了 Polyak步长在 (L_0, L_1)-平滑条件下的特性。



(a) Gradient Descent (b) Clipped Gradient Descent (c) Polyak Stepsize

(d) DecSPS (Orvieto et al., 2022) (e) AdaSPS (Jiang and Stich, 2023) (f) Inexact Polyak Stepsize

Figure 1: Convergence behaviors of various methods with the synthetic function.



(a) LSTM (b) Nano-GPT (c) T5

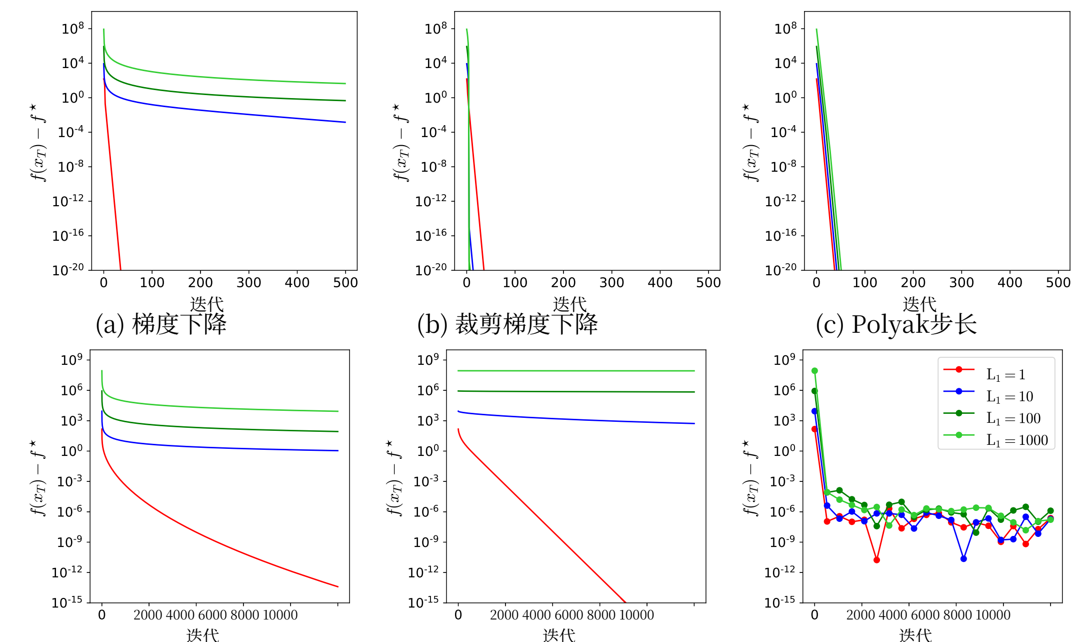
Figure 2: The final test loss with various hyperparameter settings. For T5, the results of DecSPS and AdaSPS were omitted because their final test loss was much larger than the others, as shown in Fig. 4. Furthermore, the results of SGD were also omitted when the final test loss became nan or infinity.

6.2 Neural networks

Setting: Next, we evaluated Inexact Polyak Stepsize using LSTM, Nano-GPT[2], and T5 (Nawrot, 2023). For LSTM, Nano-GPT, and T5, we used the Penn Treebank, Shakespeare, and C4 as training datasets, respectively. For SGD and Clipped SGD, we tuned the stepsize and gradient clipping threshold on validation datasets. For Polyak stepsize, we showed the results when we set $f^\star$ to zero. For Inexact Polyak Stepsize, Theorem 4 requires the selection of the best parameters. However, we do not need to choose this for neural networks because the parameters only reach the stationary point and do not reach the global minima. See Sec. D for the detailed training configuration. For all experiments, we repeated with three different seed values and reported the average.

Results: Figure 4 shows the loss curves, and Fig. 2 shows the final test losses for various hyperparameters. The results indicate that Inexact Polyak Stepsize consistently outperform DecSPS and AdaSPS for all neural network architectures. Although DoG performed the best for LSTM among the parameter-free methods, the training behavior of DoG was very unstable for Nano-GPT, and the loss values were much higher than those of the other methods. Similar to DoG, Polyak stepsize outperformed all parameter-free methods for T5, but the loss values of Polyak stepsize diverged for LSTM and Nano-GPT. Thus, Inexact Polyak Stepsize can consistently succeed in training models for all neural network architectures.

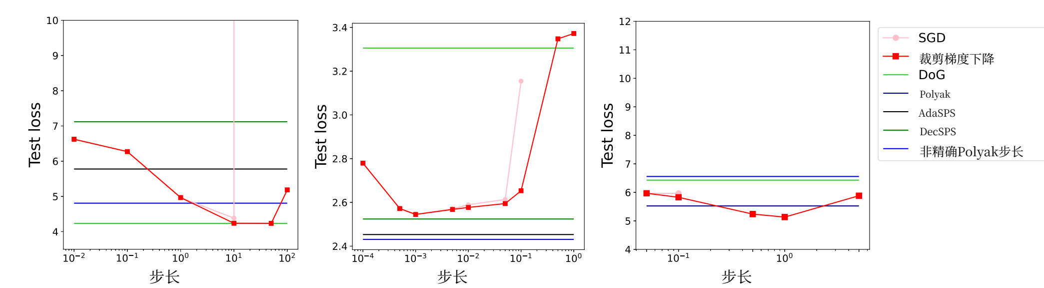
[2]`https://github.com/karpathy/nanoGPT`



(a) 梯度下降 (b) 裁剪梯度下降 (c) Polyak步长

(d) DecSPS (Orvieto等人, 2022) (e) AdaSPS (Jiang和Stich, 2023) (f) 非精确Polyak步长

图1：各种方法在合成函数上的收敛行为。



(a) LSTM (b) Nano-GPT (c) T5

图2：不同超参数设置下的最终测试损失。对于T5，DecSPS和AdaSPS的结果被省略了，因为它们的最终测试损失远大于其他方法，如图4所示。此外，当最终测试损失变为nan或infinity时，SGD的结果也被省略了。

6.2 神经网络

设置： 接下来，我们使用LSTM、Nano-GPT[2], 和T5评估了非精确Polyak步长（Nawrot，2023）。对于LSTM、Nano-GPT和T5，我们分别使用宾夕法尼亚树库、莎士比亚和C4作为训练数据集。对于随机梯度下降（SGD）和裁剪梯度下降，我们在验证数据集上调整了步长和梯度裁剪阈值。对于Polyak步长，我们展示了当我们将 $f^\star$ 设置为0时的结果。对于非精确Polyak步长，定理4要求选择最佳参数。然而，对于神经网络，我们不需要选择这个，因为参数仅达到稳定点，并未达到全局最小值。参见D节了解详细的训练配置。对于所有实验，我们使用三个不同的种子值重复进行了三次，并报告了平均值。

结果： 图4显示了损失曲线，图2显示了不同超参数的最终测试损失。结果表明，非精确Polyak步长在所有神经网络架构中始终优于DecSPS和AdaSPS。尽管差分高斯在参数无关方法中对于LSTM表现最佳，但其训练行为对Nano-GPT非常不稳定，损失值远高于其他方法。类似于差分高斯，Polyak步长在T5上优于所有参数无关方法，但Polyak步长的损失值在LSTM和Nano-GPT上发散。因此，非精确Polyak步长可以始终成功地为所有神经网络架构训练模型。

[2]https://github.com/karpathy/nanoGPT

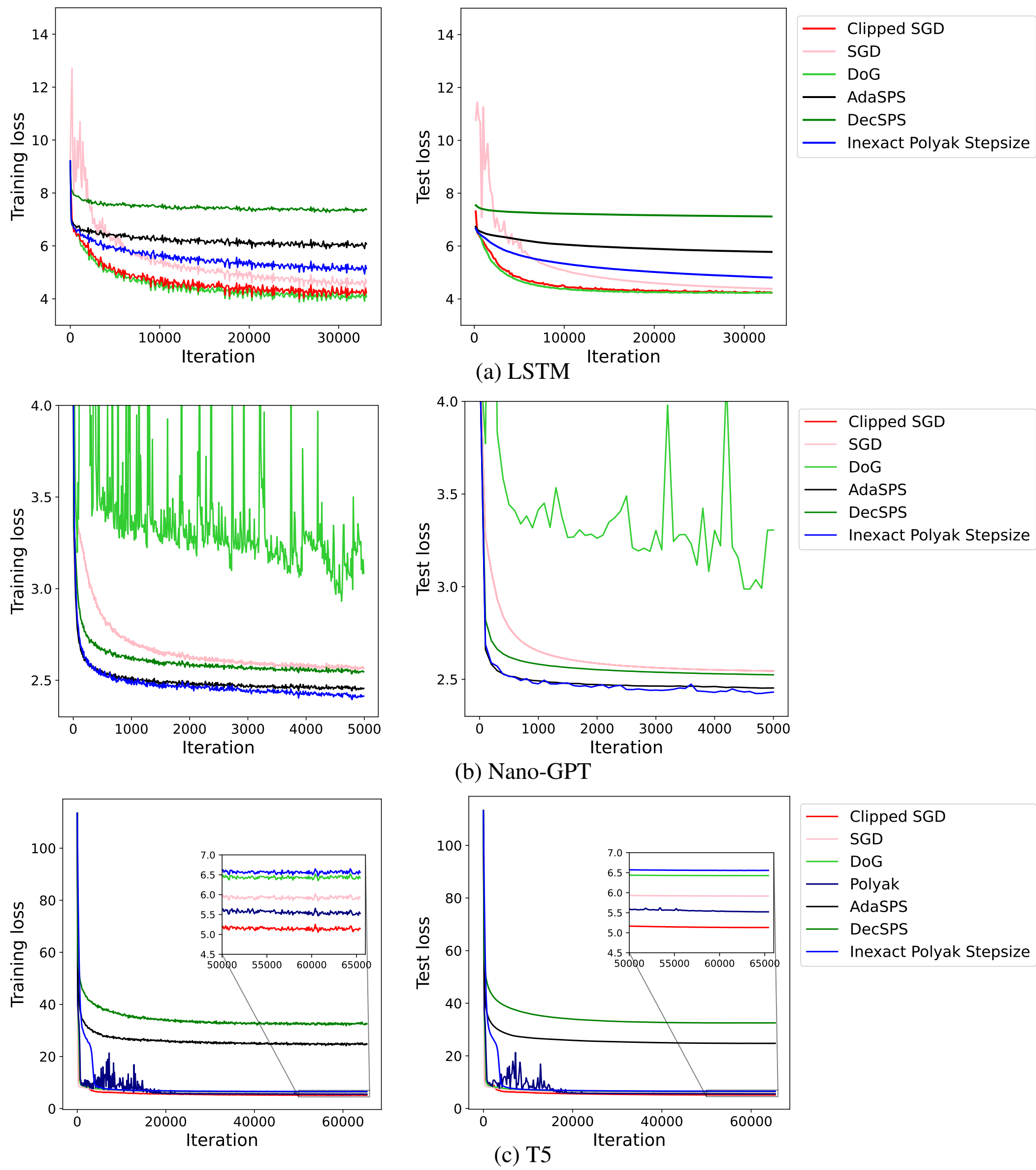


Figure 3: Loss curves for LSTM, Nano-GPT, and T5. We plotted the training loss per 100, 10, and 10 iterations for LSTM, Nano-GPT, and T5, respectively. We plotted the test loss per one epoch, 100 iterations, and 200 iterations, respectively. For LSTM and Nano-GPT, we found that Polyak stepsize does not converge, and its loss was much larger than that of other comparison methods. Thus, to make the figure easier to read, we omit the results of Polyak stepsize and provide the complete results, including Polyak stepsize in Sec. E.

7 Conclusion

In this study, we proposed Inexact Polyak Stepsize, which converges to the optimal solution without hyperparameter tuning at the convergence rate that is asymptotically independent of L under (L_0, L_1)-smoothness. Specifically, we first provided the novel convergence rate of Polyak stepsize under (L_0, L_1)-smoothness, revealing that Polyak stepsize can achieve exactly the same convergence rate as clipped gradient descent. Although Polyak stepsize can improve the convergence under (L_0, L_1)-smoothness, Polyak stepsize requires the minimum loss value, which is a problem-specific parameter. Then, we proposed Inexact Polyak Stepsize, which removes the problem-specific parameter from Polyak stepsize without losing the property of asymptotic independence of L under (L_0, L_1)-smoothness. We numerically validated our convergence results and demonstrated the effectiveness of Inexact Polyak Stepsize.

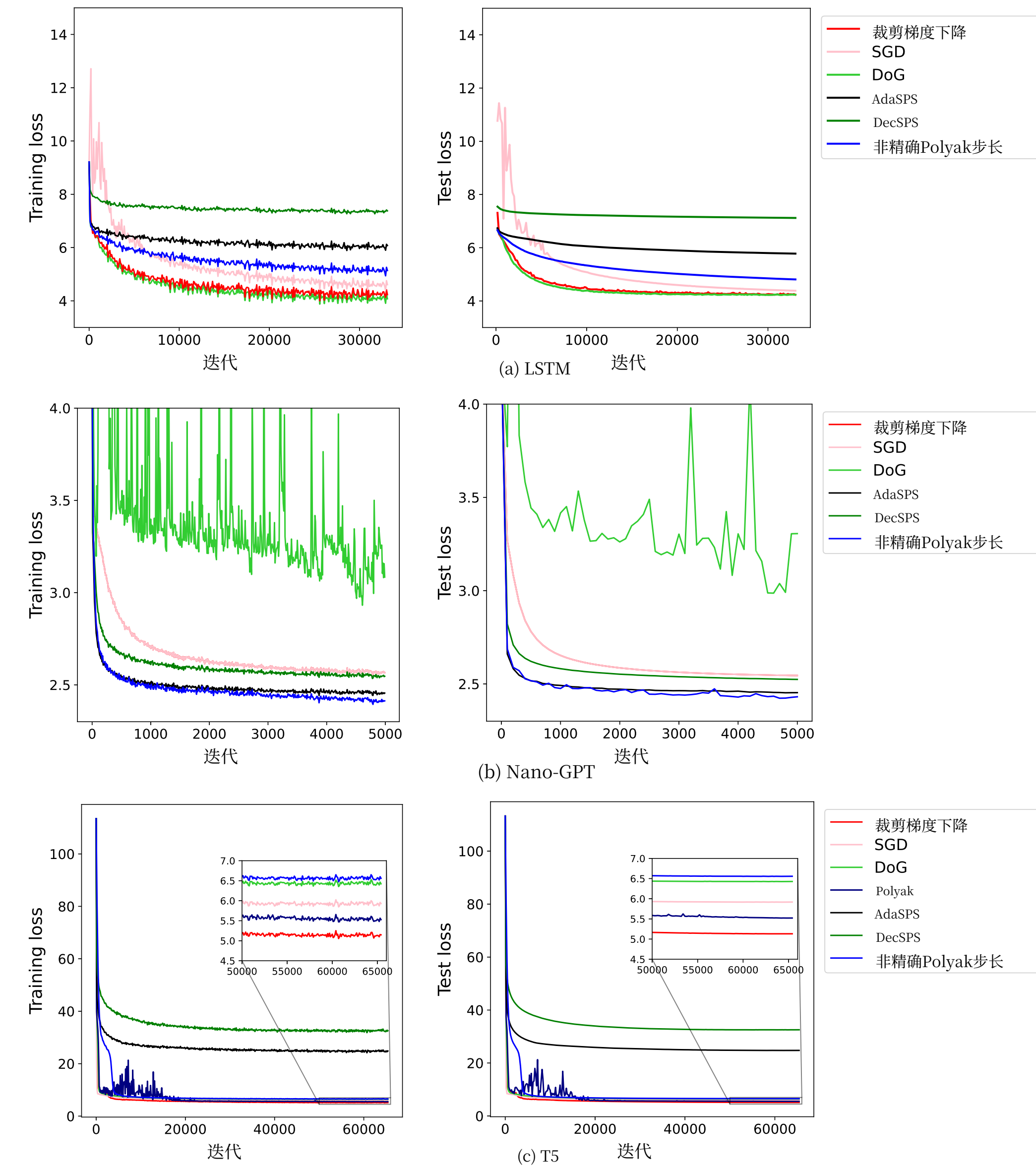


图3：LSTM、Nano-GPT和T5的损失曲线。我们分别对LSTM、Nano-GPT和T5绘制了每100次、10次和10次迭代的训练损失。我们分别绘制了每1个epoch、100次迭代和200次迭代的测试损失。对于LSTM和Nano-GPT，我们发现Polyak步长不收敛，其损失远大于其他对比方法的损失。因此，为了使图表更易读，我们省略了Polyak步长的结果，并在E节提供了完整结果，包括Polyak步长。

7 结论

在本研究中，我们提出了非精确Polyak步长，它在不进行超参数调优的情况下收敛到最优解，其收敛速度在(L_0, L_1)-平滑条件下渐近独立于 L 。具体而言，我们首先提供了Polyak步长在(L_0, L_1)-平滑条件下的新收敛速度，揭示Polyak步长可以实现与裁剪梯度下降完全相同的收敛速度。尽管Polyak步长可以在(L_0, L_1)-平滑条件下提高收敛速度，但Polyak步长需要最小损失值，这是一个特定问题的参数。然后，我们提出了非精确Polyak步长，它去除了Polyak步长中的特定问题参数，同时保留了 L 在(L_0, L_1)-平滑条件下的渐近独立性。我们通过数值验证了我们的收敛结果，并展示了非精确Polyak步长的有效性。

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11

A Proof of Theorem 4

Lemma 1. *If Assumption 1 holds, the following holds for any $\boldsymbol{x} \in \mathbb{R}^d$:*

$$\frac{1}{2L}\|\nabla f(\boldsymbol{x})\|^2 \leq f(\boldsymbol{x}) - f(\boldsymbol{x}^\star). \tag{16}$$

Proof. See Lemma 2.28 in (Garrigos and Gower, 2023). □

Lemma 2. *If Assumption 2 holds, the following holds for any $\boldsymbol{x} \in \mathbb{R}^d$:*

$$\frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x})\|)}\|\nabla f(\boldsymbol{x})\|^2 \leq f(\boldsymbol{x}) - f(\boldsymbol{x}^\star). \tag{17}$$

Proof. See Lemma A.2 in (Koloskova et al., 2023). □

Lemma 3. *Assume that f is convex and Assumption 1 and 2 hold. Let T be the number of iterations and define $\tau := \arg\min_{0 \leq t \leq T-1} f(\boldsymbol{x}_t)$. Then, gradient descent with Polyak stepsize Eq. (7) satisfies:*

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{64LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}.$$

Proof. We have

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &= \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t\langle \nabla f(\boldsymbol{x}_t), \boldsymbol{x}_t - \boldsymbol{x}^\star\rangle + \eta_t^2\|\nabla f(\boldsymbol{x})\|^2 \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) + \eta_t^2\|\nabla f(\boldsymbol{x})\|^2,\end{aligned}$$

where we use the convexity of f in the inequality.

Case when $\|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}$: Substituting the stepsize, we get

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|^2}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{4L_0}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)),\end{aligned}$$

where we use Lemma 2 in the second inequality. Unrolling the above inequality, we obtain

$$f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star) \leq 4L_0\left(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2\right).$$

Case when $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$: Substituting the stepsize, we get

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star))^2}{\|\nabla f(\boldsymbol{x}_t)\|^2} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}}\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|},\end{aligned}$$

where we use Lemmas 1 in the last inequality. Then, $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$ implies

$$\frac{L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|}{2L_1\|\nabla f(\boldsymbol{x}_t)\|} < 1.$$

Thus, we get

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}}\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|^2}\frac{L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|}{2L_1} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{4L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}},\end{aligned}$$

定理4的证明

引理1。 如果假设*1*成立，则对于任何 $\boldsymbol{x} \in \mathbb{R}^d$，以下成立：

$$\frac{1}{2L}\|\nabla f(\boldsymbol{x})\|^2 \leq f(\boldsymbol{x}) - f(\boldsymbol{x}^\star). \tag{16}$$

证明。 参见(Garrigos和Gower, 2023)中的引理2.28。 □

引理2. 如果假设*2*成立，则对任意 $\boldsymbol{x} \in \mathbb{R}^d$，以下成立：

$$\frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x})\|)}\|\nabla f(\boldsymbol{x})\|^2 \leq f(\boldsymbol{x}) - f(\boldsymbol{x}^\star). \tag{17}$$

证明。 参见(Koloskova等人, 2023)中的引理A.2。 □

引理3. 假设 f 是凸的，并且假设*1*和*2*成立。令 T 为迭代次数，并定义 τ：$= \arg\min_{0 \leq t \leq T-1} f(\boldsymbol{x}_t)$。然后，使用*Polyak*步长公式*(7)*的梯度下降满足：

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{T} + \frac{64LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T^2}.$$

证明。 我们有

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &= \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t\langle \nabla f(\boldsymbol{x}_t), \boldsymbol{x}_t - \boldsymbol{x}^\star\rangle + \eta_t^2\|\nabla f(\boldsymbol{x})\|^2 \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) + \eta_t^2\|\nabla f(\boldsymbol{x})\|^2,\end{aligned}$$

我们在不等式中使用了 f 的凸性。

当 $\|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}$**时：** 代入步长，我们得到

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|^2}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{4L_0}(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)),\end{aligned}$$

在第二个不等式中，我们使用了引理2。展开上述不等式，我们得到

$$f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star) \leq 4L_0\left(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2\right).$$

当 $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$**时：** 代入步长，我们得到

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{(f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star))^2}{\|\nabla f(\boldsymbol{x}_t)\|^2} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}}\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|},\end{aligned}$$

在最后一个不等式中，我们使用了引理1。然后，$\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$ 意味着

$$\frac{L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|}{2L_1\|\nabla f(\boldsymbol{x}_t)\|} < 1.$$

因此，我们得到

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}}\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{\|\nabla f(\boldsymbol{x}_t)\|^2}\frac{L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|}{2L_1} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \frac{1}{4L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f(\boldsymbol{x}^\star)}{2L}},\end{aligned}$$

where we use Lemma 2 in the last inequality. Unrolling the above inequality and multiplying L_0 on both sides, we get

$$\frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x})-f(\boldsymbol{x}^\star)}{2L}} \leq 4L_0\left(\|\boldsymbol{x}_t-\boldsymbol{x}^\star\|^2-\|\boldsymbol{x}_{t+1}-\boldsymbol{x}^\star\|^2\right).$$

Summing the two cases: Define $\mathcal{T}_1$ and $\mathcal{T}_2$ as follows:

$$\mathcal{T}_1 \coloneqq \left\{t \middle| \|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}\right\},\quad \mathcal{T}_2 \coloneqq \left\{t \middle| \|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}\right\}.$$

We obtain

$$\sum_{t\in\mathcal{T}_1}\left(f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)\right)+\frac{L_0}{L_1}\sum_{t\in\mathcal{T}_2}\sqrt{\frac{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)}{2L}} \leq 4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2.$$

Then, the above inequality implies

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1} f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star) \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T},$$
$$\frac{1}{T}\sum_{t\in\mathcal{T}_2}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

Using $a^2 \geq 2ab-b^2$, we obtain for any $b\in\mathbb{R}$

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\left(2b\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)}-b^2\right) \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

Thus, when $b>0$, we obtain

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{2bT}+\frac{b}{2}.$$

Choosing $b=\sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}$, we get

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}.$$

Thus, we get

$$\frac{1}{T}\sum_{t=0}^{T-1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}+\frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

Defining $\tau \coloneqq \arg\min_t f(\boldsymbol{x}_t)$, we get

$$\sqrt{f(\boldsymbol{x}_\tau)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}+\frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

Squaring the both sides, and using $(a+b)^2 \leq 2a^2+2b^2$ for all $a,b\in\mathbb{R}$, we obtain

$$f(\boldsymbol{x}_\tau)-f(\boldsymbol{x}^\star) \leq \frac{8L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}+\frac{64LL_1^2\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^4}{T^2}.$$

This concludes the statement. □

其中我们在最后一个不等式中使用了引理2。展开上述不等式并在两边同时乘以 L_0，我们得到

$$\frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x})-f(\boldsymbol{x}^\star)}{2L}} \leq 4L_0\left(\|\boldsymbol{x}_t-\boldsymbol{x}^\star\|^2-\|\boldsymbol{x}_{t+1}-\boldsymbol{x}^\star\|^2\right).$$

将两种情况相加：定义 $\mathcal{T}_1$ 和 $\mathcal{T}_2$ 如下：

$$\mathcal{T}_1 \coloneqq \left\{t \middle| \|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}\right\},\quad \mathcal{T}_2 \coloneqq \left\{t \middle| \|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}\right\}.$$

我们得到

$$\sum_{t\in\mathcal{T}_1}\left(f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)\right)+\frac{L_0}{L_1}\sum_{t\in\mathcal{T}_2}\sqrt{\frac{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)}{2L}} \leq 4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2.$$

然后，上述不等式意味着

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1} f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star) \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T},$$
$$\frac{1}{T}\sum_{t\in\mathcal{T}_2}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

使用 $a^2 \geq 2ab-b^2$，我们得到对于任何 $b\in\mathbb{R}$

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\left(2b\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)}-b^2\right) \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

因此，当 $b>0$时，我们得到

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{2bT}+\frac{b}{2}.$$

选择 $b=\sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}$，我们得到

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}.$$

因此，我们得到

$$\frac{1}{T}\sum_{t=0}^{T-1}\sqrt{f(\boldsymbol{x}_t)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}+\frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

定义 $\tau \coloneqq \arg\min_t f(\boldsymbol{x}_t)$，我们得到

$$\sqrt{f(\boldsymbol{x}_\tau)-f(\boldsymbol{x}^\star)} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}}+\frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}.$$

对两边平方，并使用 $(a+b)^2 \leq 2a^2+2b^2$ 对所有 $a,b\in\mathbb{R}$，我们得到

$$f(\boldsymbol{x}_\tau)-f(\boldsymbol{x}^\star) \leq \frac{8L_0\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^2}{T}+\frac{64LL_1^2\|\boldsymbol{x}_0-\boldsymbol{x}^\star\|^4}{T^2}.$$

这结束了陈述。 □

13

B Proof of Theorem 5

Lemma 4. *Assume that f is convex and Assumptions 1 and 2 hold. Let T be the number of iterations and define $\tau := \arg\min_{0\leq t\leq T-1} f(\boldsymbol{x}_t)$. If $f(\boldsymbol{x}_t) - f^\star \geq \frac{\sigma^2}{\sqrt{T}}$ for all t, then gradient descent with stepsize Eq. (14) satisfies:*

$$f(\boldsymbol{x}_\tau) - f^\star \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128L_1^2L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4L}{L_0^2T}.$$

where $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$ and $\sigma^2 := f^\star - l^\star$.

Proof. By the convexity of f, we have

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &= \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t\langle\nabla f(\boldsymbol{x}_t), \boldsymbol{x}_t - \boldsymbol{x}^\star\rangle + \eta_t^2\|\nabla f(\boldsymbol{x}_t)\|^2 \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f^\star) + \eta_t^2\|\nabla f(\boldsymbol{x}_t)\|^2.\end{aligned}$$

Substituting the stepsize Eq. (14), we get

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t}{\sqrt{T}}(f(\boldsymbol{x}_t) - l^\star) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \eta_t(2 - \frac{1}{\sqrt{T}})(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t\sigma^2}{\sqrt{T}} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \eta_t(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t\sigma^2}{\sqrt{T}}, \end{aligned} \tag{18}$$

where we use $T \geq 1$ in the last inequality. Unrolling the above inequality and dividing by η_t, we obtain

$$f(\boldsymbol{x}_t) - f^\star \leq \frac{\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2}{\eta_t} + \frac{\sigma^2}{\sqrt{T}}. \tag{19}$$

Case when $\|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}$: From $f(\boldsymbol{x}_t) - f^\star \geq \frac{\sigma^2}{\sqrt{T}}$ and Eq. (18), we obtain

$$\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 \geq 0. \tag{20}$$

Thus, we get

$$\begin{aligned}f(\boldsymbol{x}_t) - f^\star &\leq 2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}} \\ &\leq 4L_0\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}},\end{aligned}$$

where we use $f^\star \geq l^\star$ and Lemma 2 for the first inequality and use $\|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}$ for the last inequality.

Case when $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$: From Lemma 2, we have

$$\eta_t \geq \frac{f(\boldsymbol{x}_t) - f^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2} \geq \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)\sqrt{T}}.$$

Then, we obtain

$$\eta_t \geq \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|\sqrt{T}} \geq \frac{1}{4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}},$$

where we use $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$ for the first inequality, and Lemma 1 for the last inequality. Combining Eqs. (19) and (20), we obtain

$$f(\boldsymbol{x}_t) - f^\star \leq 4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}}.$$

B 定理5的证明

引理 4。 假设 f 是凸的，并且假设 *1* 和假设 *2* 成立。令 T 为迭代次数，并定义 τ: $=\arg\min_{0\leq t\leq T-1} f(\boldsymbol{x}_t)$。如果 $f(\boldsymbol{x}_t) - f^\star \geq \frac{\sigma^2}{\sqrt{T}}$ 对所有 t 成立，则步长为等式 *(14)* 的梯度下降满足:

$$f(\boldsymbol{x}_\tau) - f^\star \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128L_1^2L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4L}{L_0^2T}.$$

其中 $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$ 并且 $\sigma^2 := f^\star - l^\star$。

证明。 由于 f的凸性，我们有

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &= \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t\langle\nabla f(\boldsymbol{x}_t), \boldsymbol{x}_t - \boldsymbol{x}^\star\rangle + \eta_t^2\|\nabla f(\boldsymbol{x}_t)\|^2 \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f^\star) + \eta_t^2\|\nabla f(\boldsymbol{x}_t)\|^2.\end{aligned}$$

代入步长公式(14)，我们得到

$$\begin{aligned}\|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - 2\eta_t(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t}{\sqrt{T}}(f(\boldsymbol{x}_t) - l^\star) \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \eta_t(2 - \frac{1}{\sqrt{T}})(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t\sigma^2}{\sqrt{T}} \\ &\leq \|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \eta_t(f(\boldsymbol{x}_t) - f^\star) + \frac{\eta_t\sigma^2}{\sqrt{T}}, \end{aligned} \tag{18}$$

其中我们在最后一个不等式中使用了 $T \geq 1$ 。展开上述不等式并除以 η_t，我们得到

$$f(\boldsymbol{x}_t) - f^\star \leq \frac{\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2}{\eta_t} + \frac{\sigma^2}{\sqrt{T}}. \tag{19}$$

当()时： 根据 $f(\boldsymbol{x}_t) - f^\star \geq \frac{\sigma^2}{\sqrt{T}}$和公式(18)，我们得到

$$\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2 \geq 0. \tag{20}$$

因此，我们得到

$$\begin{aligned}f(\boldsymbol{x}_t) - f^\star &\leq 2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}} \\ &\leq 4L_0\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}},\end{aligned}$$

其中，我们使用 $f^\star \geq l^\star$ 和引理2 得到第一个不等式，并使用 $\|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}$得到最后一个不等式。

当 () 时： 根据引理2，我们有

$$\eta_t \geq \frac{f(\boldsymbol{x}_t) - f^\star}{\sqrt{T}\|\nabla f(\boldsymbol{x}_t)\|^2} \geq \frac{1}{2(L_0 + L_1\|\nabla f(\boldsymbol{x}_t)\|)\sqrt{T}}.$$

然后，我们得到

$$\eta_t \geq \frac{1}{4L_1\|\nabla f(\boldsymbol{x}_t)\|\sqrt{T}} \geq \frac{1}{4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}},$$

我们在第一个不等式中使用 $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$，在最后一个不等式中使用引理1。结合公式(19)和(20)，我们得到

$$f(\boldsymbol{x}_t) - f^\star \leq 4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}}.$$

Furthermore, from $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$ and Lemma 1, we obtain

$$\sqrt{f(\boldsymbol{x}_t) - f^\star} \geq \frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}} \geq \frac{L_0}{L_1}\sqrt{\frac{1}{2L}}.$$

Thus, we get

$$\begin{aligned} & f(\boldsymbol{x}_t) - f^\star \\ & \leq 4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{L_1\sigma^2}{L_0\sqrt{T}}\sqrt{2L(f(\boldsymbol{x}_t) - f^\star)}. \end{aligned}$$

Dividing by $\frac{L_1\sqrt{2L(f(\boldsymbol{x}_t)-f^\star)}}{L_0}$, we get

$$\frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f^\star}{2L}} \leq 4L_0\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}}.$$

Summing the two cases: Define $\mathcal{T}_1$ and $\mathcal{T}_2$ as follows:

$$\mathcal{T}_1 \coloneqq \left\{t \,\middle|\, \|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}\right\}, \quad \mathcal{T}_2 \coloneqq \left\{t \,\middle|\, \|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}\right\}.$$

We obtain

$$\frac{1}{T}\left(\sum_{t\in\mathcal{T}_1}(f(\boldsymbol{x}_t) - f^\star) + \frac{L_0}{L_1}\sum_{t\in\mathcal{T}_2}\sqrt{\frac{f(\boldsymbol{x}) - f^\star}{2L}}\right) \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}.$$

The above inequality implies that

$$\begin{aligned} \frac{1}{T}\sum_{t\in\mathcal{T}_1}(f(\boldsymbol{x}_t) - f^\star) &\leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}, \\ \frac{1}{T}\sum_{t\in\mathcal{T}_2}\sqrt{f(\boldsymbol{x}) - f^\star} &\leq \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}. \end{aligned}$$

Using $a^2 \geq 2ab - b^2$, we obtain for any $b \in \mathbb{R}$

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\left(2b\sqrt{f(\boldsymbol{x}_t) - f^\star} - b^2\right) \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}.$$

Thus, when $b > 0$, we obtain

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{2b\sqrt{T}} + \frac{b}{2}.$$

Choosing $b = \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}}$, we get

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}}.$$

Thus, we get

$$\frac{1}{T}\sum_{t=0}^{T-1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}} + \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}.$$

Defining $\tau \coloneqq \arg\min_t f(\boldsymbol{x}_t)$, we get

$$\sqrt{f(\boldsymbol{x}_\tau) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}} + \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}.$$

Squaring the both sides, and using $(a+b)^2 \leq 2a^2 + 2b^2$ for all $a, b \in \mathbb{R}$, we obtain

$$f(\boldsymbol{x}_\tau) - f^\star \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128L_1^2L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4L}{L_0^2T}.$$

This concludes the statement. □

此外，从 $\|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}$ 和引理1，我们得到

$$\sqrt{f(\boldsymbol{x}_t) - f^\star} \geq \frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f^\star}{\|\nabla f(\boldsymbol{x}_t)\|^2}} \geq \frac{L_0}{L_1}\sqrt{\frac{1}{2L}}.$$

因此，我们得到

$$\begin{aligned} & f(\boldsymbol{x}_t) - f^\star \\ & \leq 4L_1\sqrt{2LT(f(\boldsymbol{x}_t) - f^\star)}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{L_1\sigma^2}{L_0\sqrt{T}}\sqrt{2L(f(\boldsymbol{x}_t) - f^\star)}. \end{aligned}$$

除以 $L_1\sqrt{2L(f(\boldsymbol{x}_t)-f^\star)\ L_0}$，我们得到

$$\frac{L_0}{L_1}\sqrt{\frac{f(\boldsymbol{x}_t) - f^\star}{2L}} \leq 4L_0\sqrt{T}(\|\boldsymbol{x}_t - \boldsymbol{x}^\star\|^2 - \|\boldsymbol{x}_{t+1} - \boldsymbol{x}^\star\|^2) + \frac{\sigma^2}{\sqrt{T}}.$$

将两种情况相加： 定义 $\mathcal{T}_1$ 和 $\mathcal{T}_2$ 如下：

$$\mathcal{T}_1 \coloneqq \left\{t \,\middle|\, \|\nabla f(\boldsymbol{x}_t)\| \leq \frac{L_0}{L_1}\right\}, \quad \mathcal{T}_2 \coloneqq \left\{t \,\middle|\, \|\nabla f(\boldsymbol{x}_t)\| > \frac{L_0}{L_1}\right\}.$$

我们得到

$$\frac{1}{T}\left(\sum_{t\in\mathcal{T}_1}(f(\boldsymbol{x}_t) - f^\star) + \frac{L_0}{L_1}\sum_{t\in\mathcal{T}_2}\sqrt{\frac{f(\boldsymbol{x}) - f^\star}{2L}}\right) \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}.$$

上述不等式意味着

$$\begin{aligned} \frac{1}{T}\sum_{t\in\mathcal{T}_1}(f(\boldsymbol{x}_t) - f^\star) &\leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}, \\ \frac{1}{T}\sum_{t\in\mathcal{T}_2}\sqrt{f(\boldsymbol{x}) - f^\star} &\leq \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}. \end{aligned}$$

利用 $a^2 \geq 2ab - b^2$，我们得到对于任何 $b \in \mathbb{R}$

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\left(2b\sqrt{f(\boldsymbol{x}_t) - f^\star} - b^2\right) \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}.$$

因此，当 $b > 0$时，我们得到

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{2b\sqrt{T}} + \frac{b}{2}.$$

选择 $b = \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}}$，我们得到

$$\frac{1}{T}\sum_{t\in\mathcal{T}_1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}}.$$

因此，我们得到

$$\frac{1}{T}\sum_{t=0}^{T-1}\sqrt{f(\boldsymbol{x}_t) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}} + \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}.$$

定义 $\tau \coloneqq \arg\min_t f(\boldsymbol{x}_t)$，我们得到

$$\sqrt{f(\boldsymbol{x}_\tau) - f^\star} \leq \sqrt{\frac{4L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}}} + \frac{4L_1\sqrt{2L}\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2}{\sqrt{T}} + \frac{L_1\sigma^2\sqrt{2L}}{L_0\sqrt{T}}.$$

对两边平方，并使用$(a+b)^2 \leq 2a^2 + 2b^2$ 对所有 $a, b \in \mathbb{R}$，我们得到

$$f(\boldsymbol{x}_\tau) - f^\star \leq \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128L_1^2L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4L}{L_0^2T}.$$

这结束了陈述。□

15

Lemma 5. *Assume that f is convex and Assumptions 1 and 2 hold. Let T be the number of iterations and define $\tau := \arg\min_{0\le t\le T-1} f(\boldsymbol{x}_t)$. Then, gradient descent with stepsize Eq. (14) satisfies:*

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \le \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2 L\sigma^4}{L_0^2 T}\right), \tag{21}$$

where $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$ and $\sigma^2 := f^\star - l^\star$.

Proof. If there exists t such that $f(\boldsymbol{x}_t) - f^\star < \frac{\sigma^2}{\sqrt{T}}$, we have

$$f(\boldsymbol{x}_\tau) - f^\star \le f(\boldsymbol{x}_t) - f^\star < \frac{\sigma^2}{\sqrt{T}}.$$

Then, if $f(\boldsymbol{x}_t) - f^\star \ge \frac{\sigma^2}{\sqrt{T}}$ for all t, Lemma 4 shows that

$$f(\boldsymbol{x}_\tau) - f^\star \le \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128 L_1^2 L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4 L}{L_0^2 T}.$$

By combining the above two cases, we have the desired statement. □

C Additional theoretical result

Lemma 6. *Let f be a function such that $\|\nabla^2 f(\boldsymbol{x})\| \le L_0 + L_1\|\nabla f(\boldsymbol{x})\|$ holds for any $\boldsymbol{x}$. For any $\boldsymbol{x}, \boldsymbol{y}$ such that $\|\boldsymbol{x} - \boldsymbol{y}\| \le \frac{1}{L_0}$, we have*

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \le 2(L_0 + L_1\|\nabla f(\boldsymbol{x})\|)\|\boldsymbol{x} - \boldsymbol{y}\|.$$

Proof. See Lemma A.2 in (Zhang et al., 2020a). □

Proposition 3. *For any $L_0 \ge 0$ and $L_1 \ge 0$, $f(x) := \frac{L_0 L_1^2}{72}x^4 + \frac{L_0}{4}x^2$ is (L_0, L_1)-smooth.*

Proof. Since $f(x)$ is twice differentiable, we have

$$|\nabla^2 f(x)| = \frac{L_0 L_1^2}{6}x^2 + \frac{L_0}{2}.$$

Using $\frac{L_1}{6}x^2 + \frac{3}{2L_1} \ge |x|$, we obtain

$$\begin{aligned}|\nabla^2 f(x)| &\le \frac{L_0 L_1^2}{6}\left(\frac{L_1}{6}x^2 + \frac{3}{2L_1}\right)|x| + L_0 \\ &= \frac{L_1}{2}\left|\frac{L_0 L_1^2}{18}x^3 + \frac{L_0}{2}x\right| + \frac{L_0}{2} \\ &= \frac{L_1}{2}|\nabla f(x)| + \frac{L_0}{2}.\end{aligned}$$

From Lemma 6, we have the desired statement. □

引理 5。假设 f 是凸的，并且假设 *1* 和 *2* 成立。令 T 为迭代次数，并定义 $\tau: = \arg\min_{0\le t\le T-1} f(\boldsymbol{x}_t)$。然后，步长为等式 *(14)* 的梯度下降满足：

$$f(\boldsymbol{x}_\tau) - f(\boldsymbol{x}^\star) \le \mathcal{O}\left(\frac{L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + \sigma^2}{\sqrt{T}} + \frac{LL_1^2\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{L_1^2 L\sigma^4}{L_0^2 T}\right), \tag{21}$$

在哪里 $\boldsymbol{x}^\star := \arg\min_{\boldsymbol{x}} f(\boldsymbol{x})$ 并且 $\sigma^2 := f^\star - l^\star$。

证明。如果存在 t 使得 $f(\boldsymbol{x}_t) - f^\star < \frac{\sigma^2}{\sqrt{T}}$，我们就有了

$$f(\boldsymbol{x}_\tau) - f^\star \le f(\boldsymbol{x}_t) - f^\star < \frac{\sigma^2}{\sqrt{T}}.$$

然后，如果 $f(\boldsymbol{x}_t) - f^\star \ge \frac{\sigma^2}{\sqrt{T}}$ 对所有 t 成立，引理4表明

$$f(\boldsymbol{x}_\tau) - f^\star \le \frac{8L_0\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^2 + 2\sigma^2}{\sqrt{T}} + \frac{128 L_1^2 L\|\boldsymbol{x}_0 - \boldsymbol{x}^\star\|^4}{T} + \frac{8L_1^2\sigma^4 L}{L_0^2 T}.$$

通过结合上述两种情况，我们得到了所需的结果。□

C 额外的理论结果

引理6。设 f 是一个函数，使得 $\|\nabla^2 f(\boldsymbol{x})\| \le L_0 + L_1\|\nabla f(\boldsymbol{x})\|$ 对任何 $\boldsymbol{x}$ 成立。对于任何 $\boldsymbol{x}$, $\boldsymbol{y}$ 满足 $\|\boldsymbol{x} - \boldsymbol{y}\| \le \frac{1}{L_0}$，我们有

$$\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \le 2(L_0 + L_1\|\nabla f(\boldsymbol{x})\|)\|\boldsymbol{x} - \boldsymbol{y}\|.$$

证明。参见(Zhang et al., 2020a)中的引理A.2。□

命题3。对于任意 $L_0 \ge 0$ 和 $L_1 \ge 0$， $f(x) := \frac{L_0 L_1^2}{72}x^4 + \frac{L_0}{4}x^2$ 是 (L_0, L_1)-光滑的。

证明。由于 $f(x)$ 是二阶可微的，我们有

$$|\nabla^2 f(x)| = \frac{L_0 L_1^2}{6}x^2 + \frac{L_0}{2}.$$

利用 $\frac{L_1}{6}x^2 + \frac{3}{2L_1} \ge |x|$，我们得到

$$\begin{aligned}|\nabla^2 f(x)| &\le \frac{L_0 L_1^2}{6}\left(\frac{L_1}{6}x^2 + \frac{3}{2L_1}\right)|x| + L_0 \\ &= \frac{L_1}{2}\left|\frac{L_0 L_1^2}{18}x^3 + \frac{L_0}{2}x\right| + \frac{L_0}{2} \\ &= \frac{L_1}{2}|\nabla f(x)| + \frac{L_0}{2}.\end{aligned}$$

从引理6中，我们得到了所需的结果。□

D Hyperparameter settings

D.1 Synthetic function

In our experiments, we ran the clipped gradient descent with the following hyperparameters and tuned the hyperparameters by grid search.

Table 2: Hyperparameter settings for clipped gradient descent.

Learning Rate	$\{1, 1.0 \times 10^{-1}, \cdots, 1.0 \times 10^{-8}\}$
Gradient Clipping Threshold	$\{0.01, 0.1, 1, 5, 10, 15, 20, \infty\}$

Table 3: Hyperparameters selected by grid search.

	Gradient Descent	Clipped Gradient Descent	
	Learning Rate	Learning Rate	Gradient Clipping Threshold
$L_1 = 1$	1.0×10^{-1}	0.1	20
$L_1 = 10$	1.0×10^{-3}	0.1	10
$L_1 = 100$	1.0×10^{-5}	0.1	10
$L_1 = 1000$	1.0×10^{-7}	0.1	10

D.2 Neural networks

In our experiments, we used the following training configuration:

- **LSTM:** `https://github.com/salesforce/awd-lstm-lm`
- **Nano-GPT:** `https://github.com/karpathy/nanoGPT`
- **T5:** `https://github.com/PiotrNawrot/nanoT5`

We ran all experiments on an A100 GPU. For Clipped SGD and SGD, we tuned the stepsize and gradient clipping threshold using the grid search. See Tables 4, 5, and 6 for detailed hyperparameter settings, and see Table 7 for the selected hyperparameters.

Table 4: Hyperparameter settings for LSTM.

Learning Rate	$\{100, 50, 10, 1, 0.1, 0.01\}$
Gradient Clipping Threshold	$\{0.5, 1, \cdots, 4.5, 5, \infty\}$
Batch Size	80

Table 5: Hyperparameter settings for Nano-GPT.

Learning Rate	$\{1, 0.5, 0.1, \cdots, 0.0005, 0.0001\}$
Gradient Clipping Threshold	$\{1, 2, \cdots, 9, 10, \infty\}$
Batch Size	64

Table 6: Hyperparameter settings for T5.

Learning Rate	$\{5.0, 1.0, 0.5, 0.1, 0.05\}$
Gradient Clipping Threshold	$\{1, 2, 3, \infty\}$
Batch Size	128

Table 7: Hyperparameters selected by grid search. Three values correspond to the selected hyperparameters for different seed values.

	Gradient Descent	Clipped Gradient Descent	
	Learning Rate	Learning Rate	Gradient Clipping Threshold
LSTM	10 / 10 / 10	10 / 50 / 50	0.5 / 1 / 0.5
Nano-GPT	0.001 / 0.001 / 0.001	0.001 / 0.001 / 0.001	∞ / 10 / 10
T5	0.1 / 0.1 / 0.05	1 / 1 / 1	2 / 2 / 2

D 超参数设置

D.1 合成函数

在我们的实验中，我们使用了以下超参数运行裁剪梯度下降，并通过网格搜索调整了超参数。

表2：裁剪梯度下降的超参数设置。

学习率	$\{1, 1.0 \times 10^{-1}, \cdots, 1.0 \times 10^{-8}\}$
梯度裁剪阈值	$\{0.01, 0.1, 1, 5, 10, 15, 20, \infty\}$

表3：网格搜索选择超参数。

	梯度下降	裁剪梯度下降	
	学习率	学习率	梯度裁剪阈值
$L_1 = 1$	1.0×10^{-1}	0.1	20
$L_1 = 10$	1.0×10^{-3}	0.1	10
$L_1 = 100$	1.0×10^{-5}	0.1	10
$L_1 = 1000$	1.0×10^{-7}	0.1	10

D.2 神经网络

在我们的实验中，我们使用了以下训练配置：

- **LSTM:**https://github.com/salesforce/awd-lstm-lm
- **Nano-GPT:**https://github.com/karpathy/nanoGPT
- **迭代次数5：**https://github.com/PiotrNawrot/nanoT5

我们在A100 GPU上运行了所有实验。对于裁剪梯度下降和随机梯度下降，我们使用网格搜索调整步长和梯度裁剪阈值。有关详细的超参数设置，请参见表4、表5和表6，有关选定的超参数，请参见表7。

表4：LSTM的超参数设置。

学习率	$\{100, 50, 10, 1, 0.1, 0.01\}$
梯度裁剪阈值	$\{0.5, 1, \cdots, 4.5, 5, \infty\}$
批大小	80

表5：Nano-G的超参数设置 PT.

学习率	$\{1, 0.5, 0.1, \cdots, 0.0005, 0.0001\}$
梯度裁剪阈值	$\{1, 2, \cdots, 9, 10, \infty\}$
批大小	64

表6：T5的超参数设置。

学习率	$\{5.0, 1.0, 0.5, 0.1, 0.05\}$
梯度裁剪阈值	$\{1, 2, 3, \infty\}$
批大小	128

表7：网格搜索选定的超参数。三个值对应于不同种子值选定的超参数。

	梯度下降	裁剪梯度下降	
	学习率	学习率	梯度裁剪阈值
LSTM	10 / 10 / 10	10 / 50 / 50	0.5 / 1 / 0.5
Nano-GPT	0.001 / 0.001 / 0.001	0.001 / 0.001 / 0.001	∞ / 10 / 10
T5	0.1 / 0.1 / 0.05	1 / 1 / 1	2 / 2 / 2

E Additional numerical evaluation

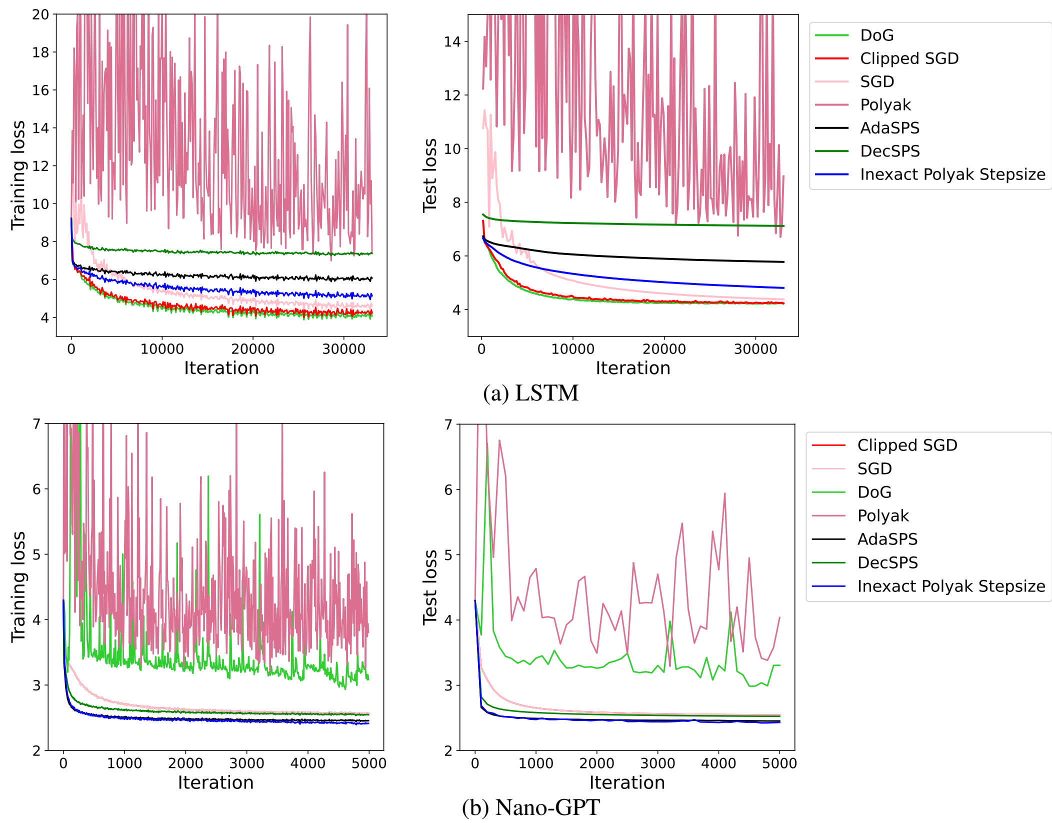


Figure 4: Loss curves for LSTM and Nano-GPT. We plotted the training loss per 100, 10, and 10 iterations for LSTM, Nano-GPT, and T5, respectively. We plotted the test loss per one epoch, 100 iterations, and 200 iterations, respectively.

E 额外的数值评估

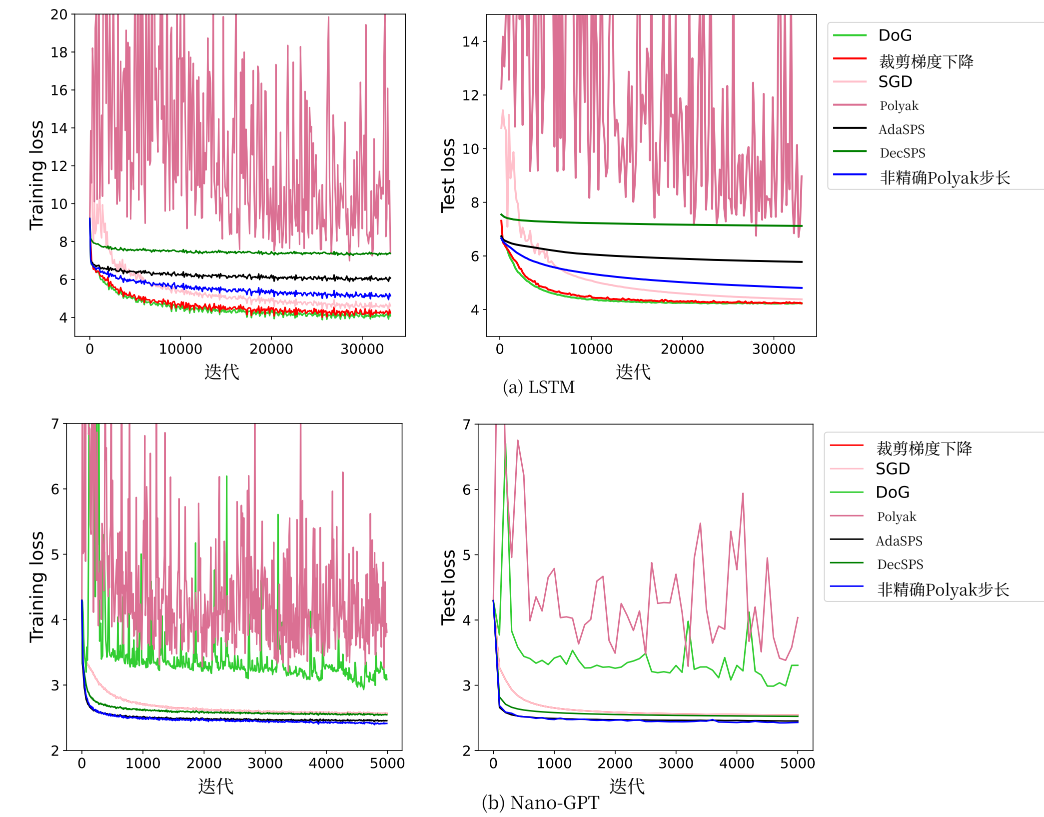
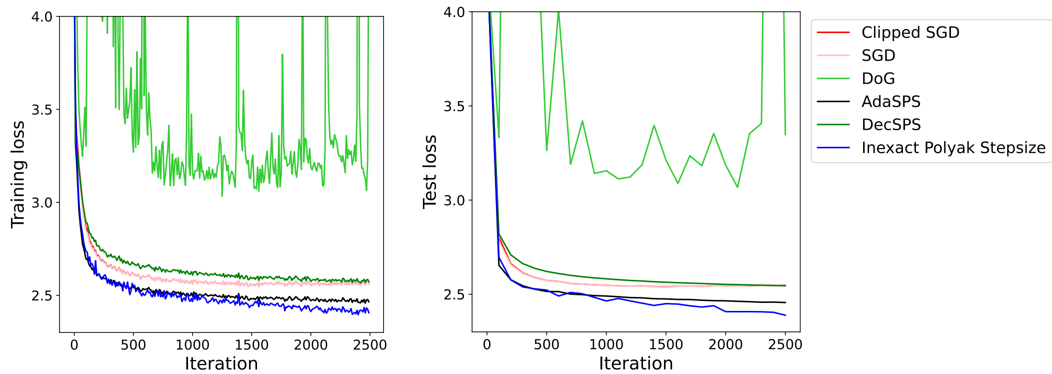
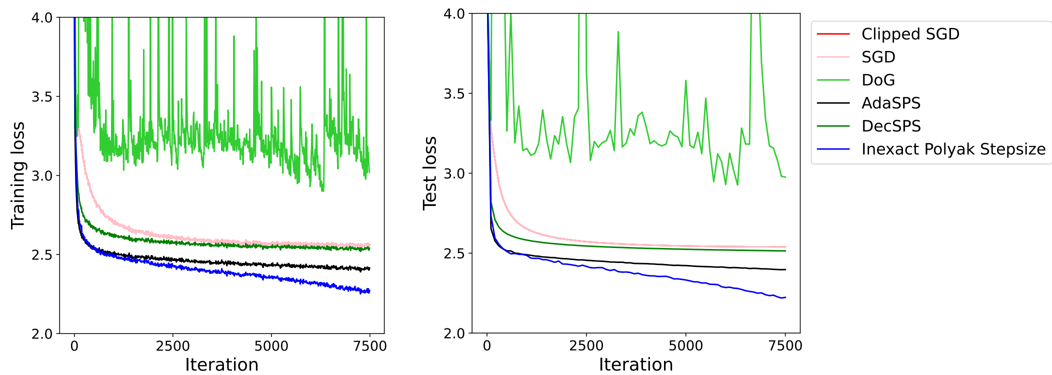


图4：LSTM和Nano-GPT的损失曲线。我们分别绘制了LSTM、Nano-GPT和T5的训练损失，单位为每100次、10次和10次迭代。我们分别绘制了测试损失，单位为每1个epoch、100次迭代和200次迭代。

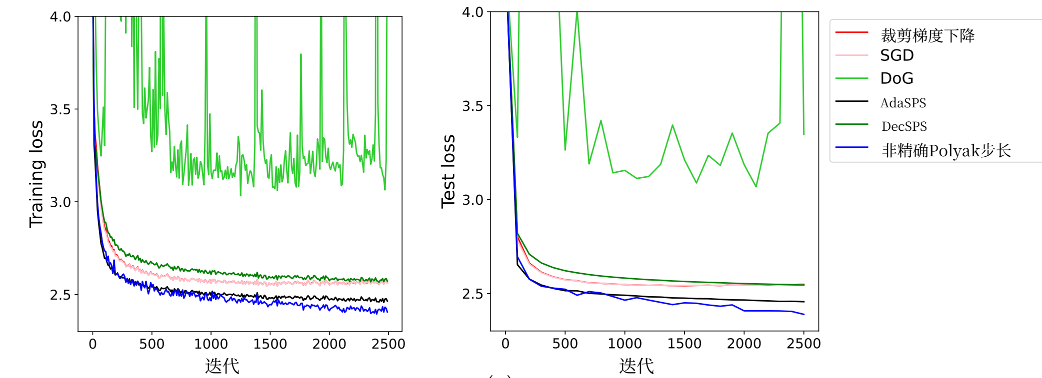


(a) $T = 2500$

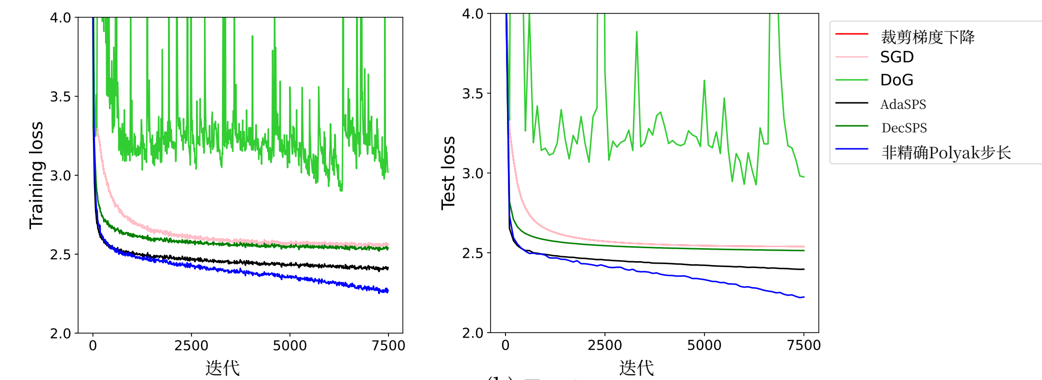


(b) $T = 7500$

Figure 5: Loss curves for Nano-GPT with different T.



(a) $T = 2500$



(b) $T = 7500$

图5：不同Nano-GPT的损失曲线。

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Our main claims are clearly discussed in Sec. 1.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: See Sec. 6.2.

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   答案： [是]

   理由：参见第6.2节。

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   问题：对于每个理论结果，论文是否提供了完整的假设集和完整的（且正确的）证明?

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Justification: All proofs are provided in Sec. A and B.

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Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

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5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

理由：所有证明都在A和B部分提供。

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回答： [是]

理由：所有训练配置和超参数设置均在D节中提供。

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Answer: [Yes]

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- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
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Answer: [Yes]

Justification: Our training configuration is provided in Sec. D.

Guidelines:

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Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: We repeated the experiments in Sec. 6.2 with three different seed values and reported the average, while we did not report error bars to make the figure easy to read.

Guidelines:

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- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).

回答： [是]

说明：我们的代码包含在补充材料中。

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答案： [是]

理由：我们的训练配置在D节中提供。

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- 实验设置应在论文核心部分以必要详细程度呈现，以便读者理解结果并做出判断。
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回答： [否]

理由：我们在第6.2节中用三个不同的种子值重复了实验，并报告了平均值，但我们没有报告误差线以使图形易于阅读。

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23

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24

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